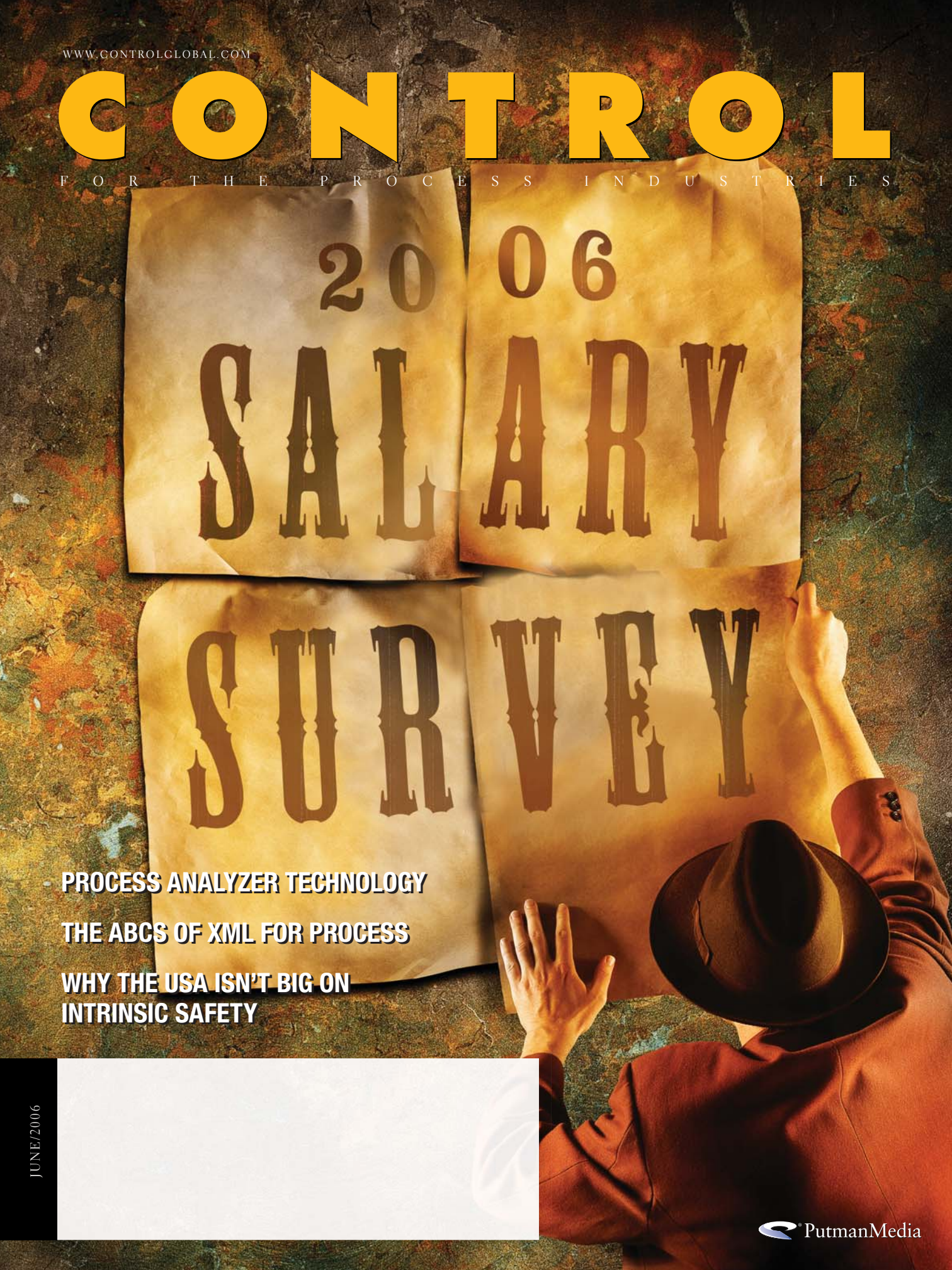
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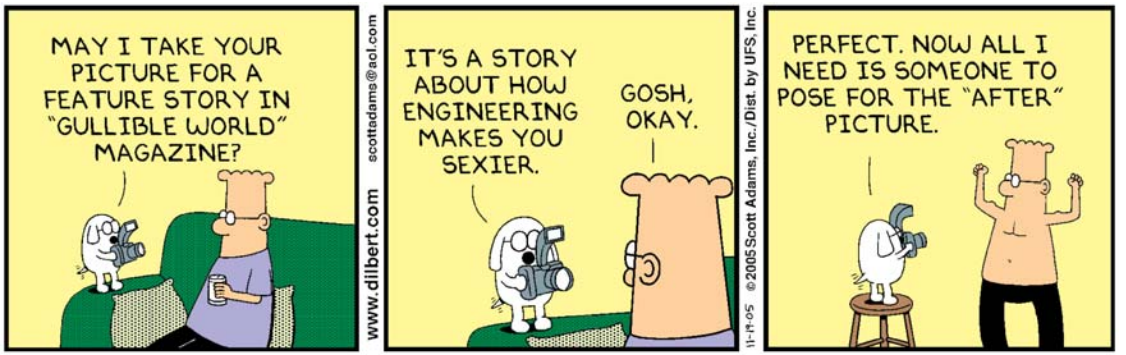
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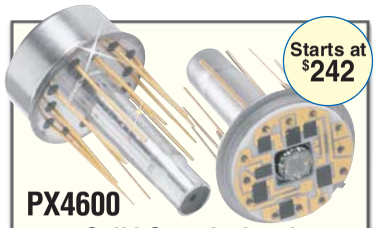
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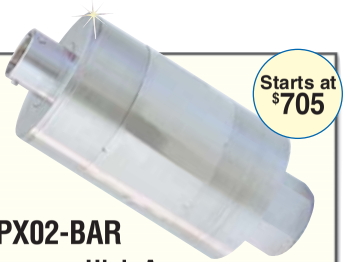
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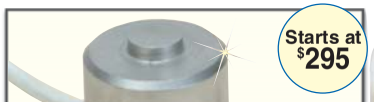
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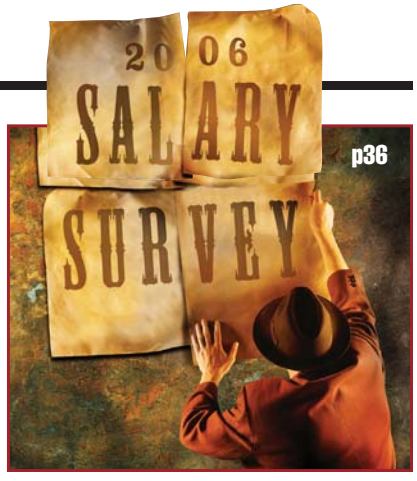
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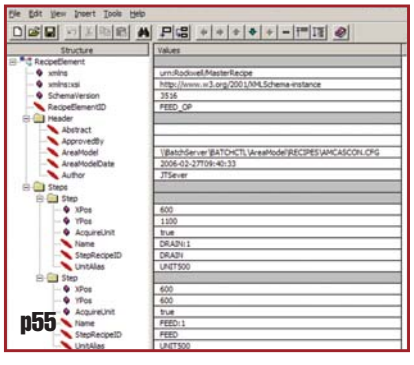
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We're seeing the end of analog transmitters and proprietary protocols.

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Chemicals & Allied Products	15,926
Food & Kindred Products	15,821
Paper & Allied Products	4,087
Primary Metal Industries	5,341
Electric, Gas & Sanitary Services	2,707
System Integrators & Engineering Design Firms	7,933
Rubber and Miscellaneous Plastic Products	5,016
Stone, Clay, Glass & Concrete products	2,580
Textile Mill Products	2,088
Petroleum Refining & Related Industries	3,361
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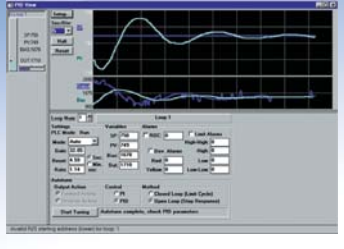


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Whither Wireless?

Dear Automation Vendors: I am on my knees. The single most important technical development issue facing process automation is wireless in the industrial plant environment, and we're in serious danger of screwing it up badly.

We all know what a mess the fieldbus standard war was, and still is. After 15 years, you're still arguing about FDT and EDDL, instead of uniting to bring the benefits of fieldbus technology to the end-users, who've replied by voting with their dollars. There are fewer than 500,000 Foundation fieldbus-enabled devices in the world, and nearly 21 million HART enabled devices. Why? Because HART transmitters all worked the same way for years before the SP50 committee got its act reasonably together.



Now comes the recent controversy over wireless standards. Everybody is pointing fingers at everybody else. End users don't really care what the standard says, except for those few on the standards committees, who represent the far right-hand side of the bell curve. They just want a standard. Any workable standard will do.

Please contrast this situation with the marketplace for fax machines. In 1980, I had a wirephoto transmitting device that sent one page every 30 minutes. By 1985, I had an optical fax machine, operatng at 300 baud, and by 1990, the current fax machine was born.

For 20 years, from 1985 to 2005, millions and millions of fax machines were sold, and sold, and sold. Why? The International Telecommunications Union's fax standard was agreed to early, and adopted by every vendor.

These vendors saw a big pot of money in the hands of end-users of their products, and they knew that the only way they would ever get their hands on it was if they cooperated, so they could compete. Other groups have developed standards that way too. It's time you all stepped up and played nice.

Unfortunately for end-users in process automation, you all have a long history of competing before you cooperate in the standards making process. One high-ranking vendor told me recently, "It's a balancing act between wanting to act in the best interest of the end users, and trying to squeeze some little competitive advantage out of the standard."

ITU had the same problem we have with developing technology, and its engineers wrote a standard that could be both forward and backward compatible. This allowed them to adopt the best available technology, and amend the standard as the technology improved. Just like ITU, we don't have the luxury of waiting.

Another big player in this game told me he doesn't think there's a market for wireless yet. He believes, as do I, that the market will happen after the standard is born, and the size of the market is going to be determined by the speed at which a standard is adopted, and actual working stuff gets in the field.

Nobody has all the answers for industrial wireless. However, you vendors have the ability to create a standard that describes best-available technology, while preserving room to maneuver for development.

The end user community isn't looking for Mr. Goodbar or a silver bullet. They want a dependable standard they can specify for wireless in their plants. Is that too much to ask?



***Walt Boyes,** editor in chief*
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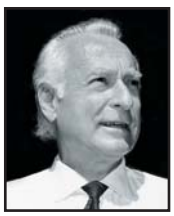
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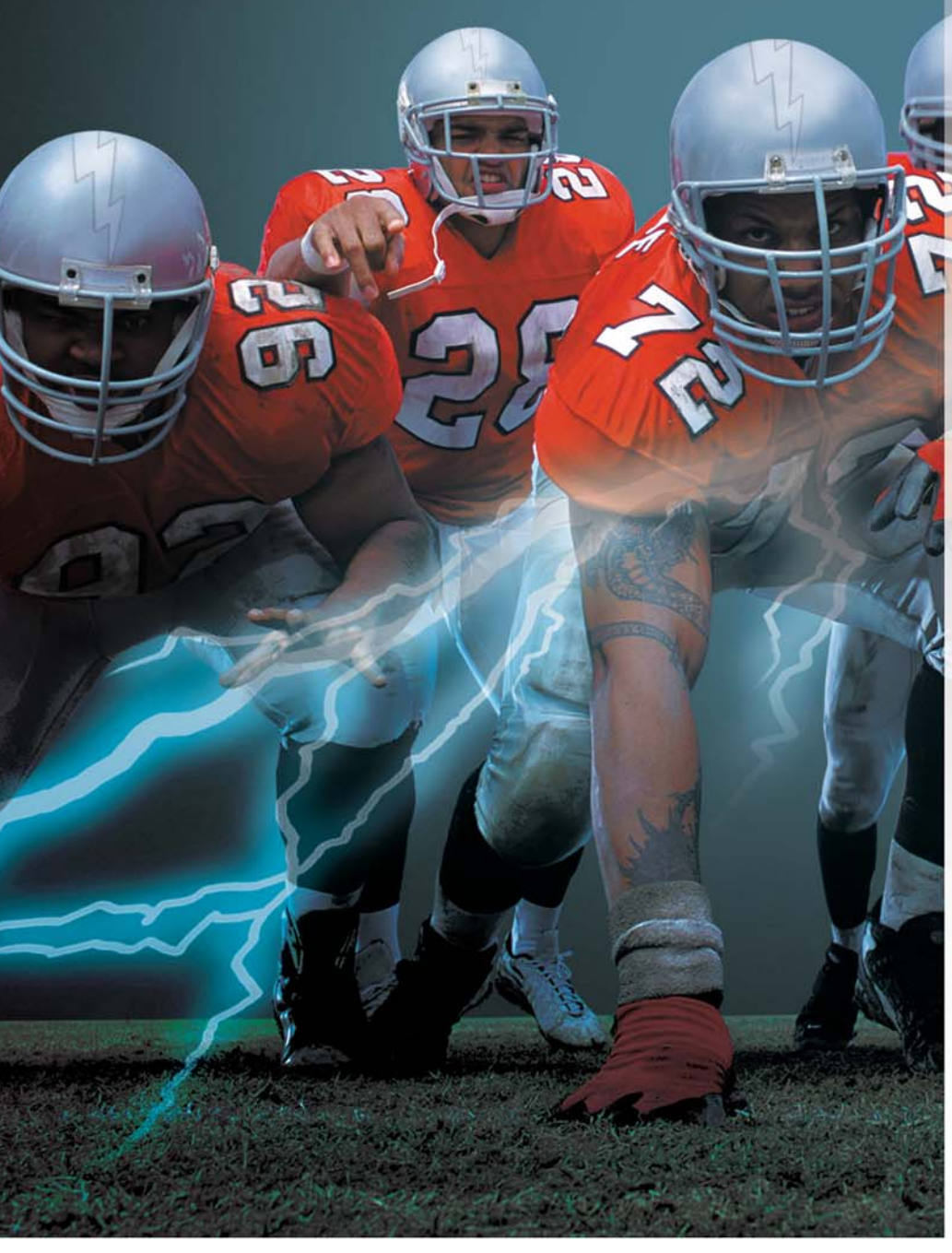
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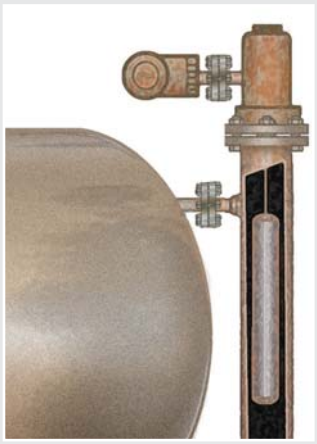
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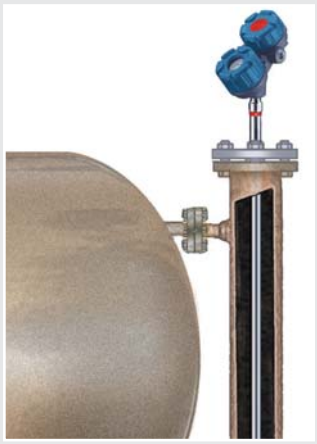
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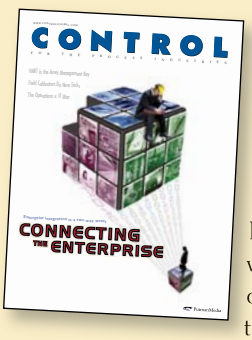
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Management the Navy Way

Very good piece ("Management Lessons from the U.S. Navy," Control Report, *Control*, March '06, p. 31), but you're preaching to the converted. I was with Chevron when *In Search of Excellence* came out. They decided we should all take the course, and Chevron should become an "excellent" company. Unfortunately, they hadn't read the book to find out that "excellence" started with the top management, and how they treat people on down the chain. Instead they just wanted people to act like those in the "excellent" companies. Needless to say the program was less than successful.



Ray Grigsby, PE
senior I&C engineer

Microsoft Factory Coming

I agree with your opinion in "Microsoft Factory Getting Ready to Take Control," Control Report, *Control*, Feb. '06, p. 35. We're already doing small and large applications with Microsoft products, and filling the gaps that exist on the HMI/SCADA front. We started using Microsoft products for HMI/SCADA in 1992, and switched to .NET for HMI/SCADA when .NET was in its beta. Microsoft Factory isn't in the future. It's already here. Unfortunately, many folks in the control industry are ignorant about it. We've been trying to educate them about it wherever we can.

Any large-scale HMI/SCADA systems we've seen implemented using non-Microsoft products are kind of a contaminated mess because the solutions consist of running multiple products from multiple software vendors.

Rakesh Verma
Parijat controlware, Inc.

Global Warming Hoax Redux

In contrast to Kirk Wishowski's letter, "More Facts, Less Venom on Global Warming," *Control*, April '06, p. 15, I clipped the articles he criticized, and posted them on our bulletin board. Scientists today seem to look into a far distant future they'll never see, and predict what it will be like. Or, they come up with outlandish theories about how the world was formed, which can never be proven or disproven, and count their theories as fact. They talk with great authority about black holes, what they are and how they were formed, but who will ever know the truth? However, when it comes to looking at the history of global warming "and cooling," they can't seem to see the nose in front of their faces. The world has been warming since the Ice Age. There have been hundreds of years of greater than average warmth and periods of greater than average cold. The big picture is that you can't look at a snapshot of 10 or 20 years, or even 100 years, and come up with a long-term trend of the world's climate. The effect of man and his machines on the world's climate is greatly exaggerated. One eruption of a volcano or asteroid impacting the earth will have a greater effect on Earth's climate than all of the machines man will ever produce. I agree with Mr. Walsh and Mr. Merritt's columns that "global warming" is a myth! We have too many scientists with too much time on their hands with no purpose for their existence other than come up with theories that can never be proven or disproven, who are taken all too seriously by the mainstream media. Thank you for printing the truth, and keep on giving voice to the other side of the issue.

Stephen Kline
electrical engineer

The Relay Logic Syndrome

Valves, sensors, actuators, etc. are comfortable tools of our trade. Today. However, while these will continue to be significant elements of control and automation solutions, several intertwining areas of explosive growth are poised to offer almost unimaginable changes and improvements, which will open up vast, though physically tiny, new control opportunities and tools. Swimming in these new seas will require very different engineering knowledge, techniques, and piece-parts.



The melding of nanotechnology, biology and medicine, information sciences and cognitive sciences (NBIC) will give automation and control devices new performance and capabilities.

The word that probably springs to mind is "nanotechnology," broadly defined as working with things sized below 100 nanometers (100 billionths of a meter). And it's true that new building blocks, such as carbon nanotubes, will offer incredible opportunities for tiny electronics (such as carbon nanotube transistors), new materials, and new venues for process control (as in a complex fluidic process implemented on one tiny chip).

However, the real changes and opportunities that lie ahead are only partially based on nanotechnology, and will blend four previously disparate fields in ways that fuel synergistic advances so quickly and so broadly that we'll wonder why we ever thought that Moore's Law of the past 35 years was considered "fast."

This blending will involve the coming together of nanotechnology, biology and medicine, information sciences and cognitive sciences (NBIC), and will certainly give traditional automation and control devices new performance and capabilities. However, even newer capabilities and challenges will appear as the boundaries blur between things living and dead and mechanical and biological and electronic items.

We have to keep up, or we'll find ourselves feeling like "relay logic" engineers, who refused to educate themselves as transistors, chips, and PLCs silenced the traditional, quiet clicking. For now, NBIC inroads are so new that there's no roadmap yet to define a self-education path. Nevertheless, we can begin to get an overview of today's laboratory and prototype NBIC efforts to help us understand the breadth and the differences in tomorrow's environments, tools, and "machines."

Superman's Muscles

The comic book hero is well known for his ultra-strong muscles, but similar artificial "muscles" might be just as useful to replace traditional electric motors in many applications. These are hydrogen-powered, artificial muscles that would be 100 times stronger than normal muscles, and able to do 100 times the work, while burning hydrogen and alcohol! They may one day power prosthetic limbs, autonomous robots, and large and tiny control devices.

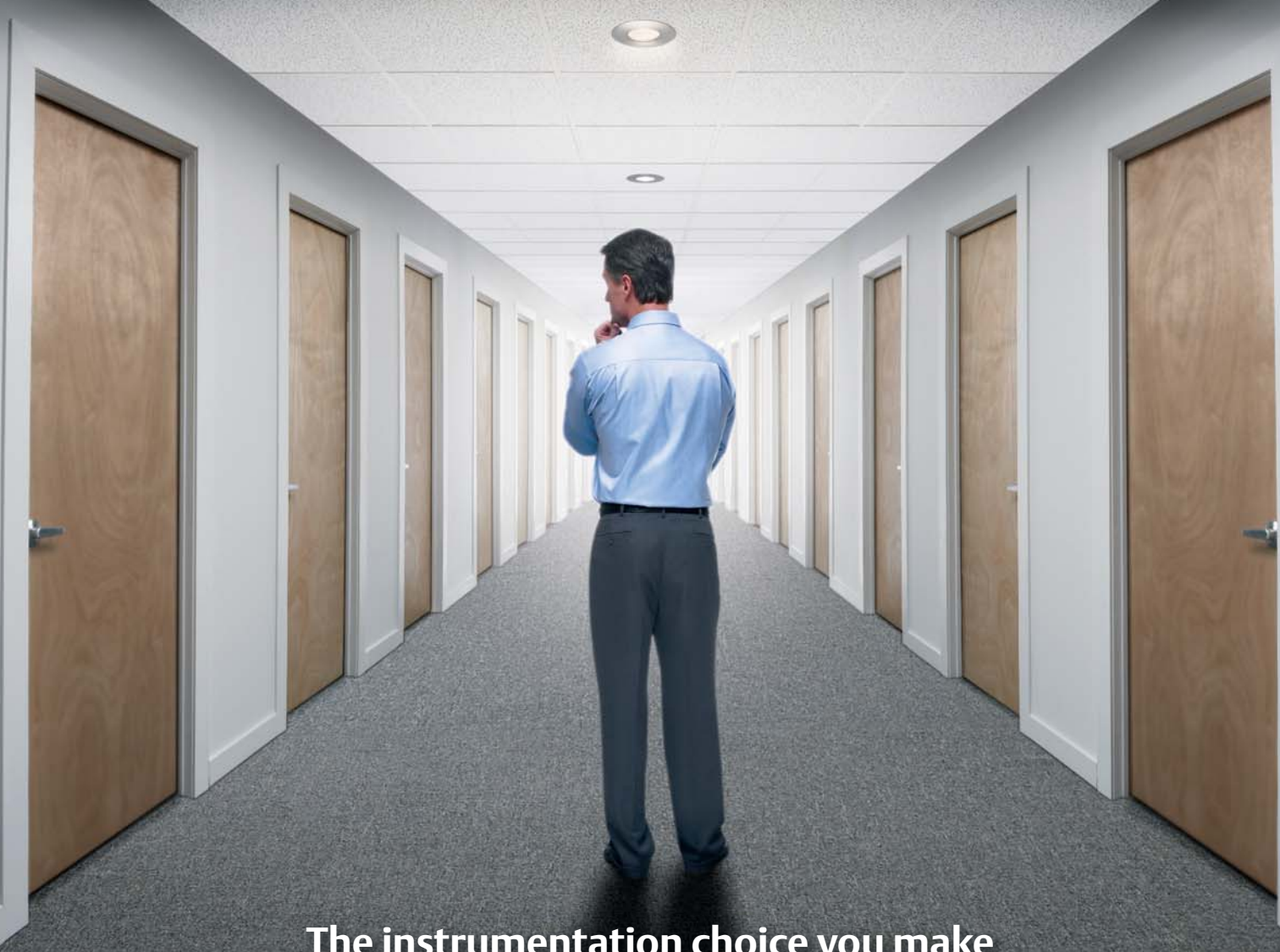
Carbon nanotubes are at the heart of this invention in several ways: in a fuel cell-type reaction to generate energy; as supercapacitors to store the energy; and, using quantum mechanical and electrostatic effects, to induce a "dimensional change," whch are the muscle-like contractions that do the work. Such building blocks will enable machines that weren't previously practical such as wings that continuously morph to optimize flight, replacement human muscles that derive their energy from the same "fuel" used by natural muscles, and many others.

Using Nemo

Pixar and Disney recently entertained us with a film about fish in the ocean, but we're moving towards a day when tiny, autonomous robots will swim through pipes and blood vessels. Many motive designs are being explored, such as molecule-sized, rotating propellers, whipping flagella like some bacteria, and even fast-moving "Push-Me-Pull-You" microswimmers. What could these be good for? Perhaps to deposit repair materials to a production surface without shutting down its process. Or carry a solvent to dissolve a "clot" in human or fluidic systems. As we become familiar with the "technology," we'll envision new approaches to old problems, and solutions to new problems we haven't dreamed of yet.

Squeezable and Strong

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which will have characteristics so astounding that many heads will shake in wonder. This is just one example of how nanotechnology is changing material characteristics at the nano and macro scales (*Nature*, Nov. 24, 2005).

More Power for Less

Most control devices, whether they're sensors or actuators, require power to do ther jobs. Yet in outoor and remote areas, power can be difficult and expensive to provide. Today, some of these needs are met by solar cell arrays, but they're notoriously expensive and inefficient, converting at best 15-20% of light into current. So, what if you could buy solar cells appraching 40-50% efficiency? Though this doublng of performance may seem unlikely, using carbon nanotubes is already leading us in this direction.

One company, AMBIT (www.ambitcorp.com), already has found a way to add about 1 billion nanotubes to traditional silicon solar cells. These tubes act as "optical antennas" to recover more ambient light energy partially by extending the light spectrum from which they can convert energy. This technology is called "nanotube-on-silicon," and it's already demonstrated an 18% efficiency improvement over traditional solar cells. This may be the beginning of a revolution in solar cell efficiency. In fact, DARPA is fundng a $54 million research project, which may use "engineered biomolecules to guide the assembly of inorganic materials" to produce cost-effective solar cell prototypes that deliver 50% efficiency by 2009. Such efficient remote power might enable sensing and control in areas that were too expensive before, and enable a decentralized energy grid as buildings become more energy self-sufficient.

All possible due to the melding of NBIC technologies, these ideas represent the tip of the iceberg of changes that are already beginning to radically alter the ways we engineer products and control processes. These changes are taking place so fast, and now the rate of change itself is accelerating because of the synergy of NBIC. So, when examining our careers it's not safe to think, "I'll just let the next generation of engineers train for this." Continuing education is the key to keeping "relay logic" syndrome at bay. Don't blink!

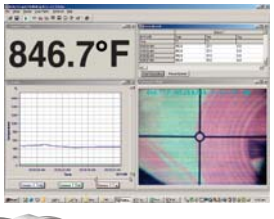
Jeffrey R. Harrow is principal technologist at The Harrow Group. He can be reached at jeff@theharrowgroup.com.

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Show Me the Installations

It appears that all of us are becoming not only more reliant on our suppliers, but also on the information they provide. Gone are the days of healthy skepticism where the proverbial "proven in use," backwards compatibility, long term support, openness, and interoperability were required before the engineer would be willing to consider installing something in the plant.

Since 1994, Foundation fieldbus (FF) has been demonstrating backwards compatibility and interoperability. After about 10 years since the introduction of the FF "check mark," Foundation fieldbus is now widely accepted as a reliable platform on which to design, operate, and control a facility, even though many people are still unsure about "control in the field," despite the fact they were doing this with pneumatics 50 years ago. When asked why they aren't using control in the field, the most common answer is that they want to see it proven elsewhere.



I may not be from Missouri, but I still want you to "show me."

It's therefore interesting that we don't hear the same degree of skepticism about FDT/DTM. Perhaps it's because people are "only" using FDT/DTM to communicate with field devices for maintenance and diagnostics, which are believed to be less important because they're "not really control." Unfortunately, because changes *can* be made to the operating conditions of a device via the maintenance system, it can no longer simply be treated separately from the control system. These two are an integrated whole.

FDT/DTM adds another layer of software and, hence, potential complexity to the system. And, with the new enhancements to EDDL, FDT/DTM is a duplication of effort to some extent. FDT/DTM can require a large effort to implement since its software needs a multi-megabyte software download of 350 Megs. Consequently, FTD/DTM can't execute in a field device alone–at least not yet. Because FDT/DTM adds its layer between the field device and the maintenance technician, it further increases the system's complications. Obviously, due to its file size, FDT/DTM must reside on a dedicated computer.

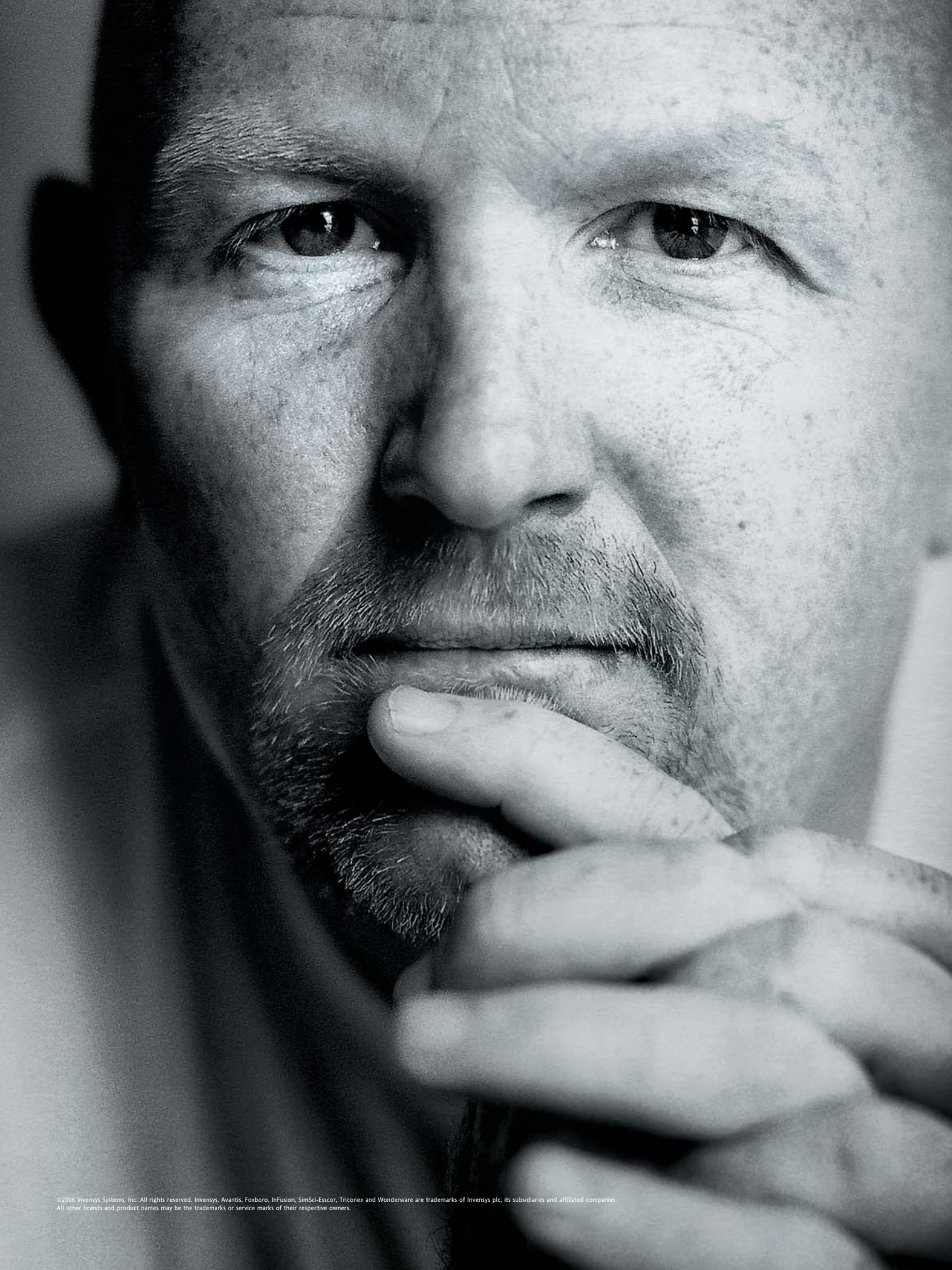
Interestingly, if you investigate the FDT-JIG website, approximately 85% of the FDT/DTM files available for download are from just two European suppliers. This isn't what one would expect from a global standard, especially one that promotes its large membership base on its other web pages. Granted FDT/DTM has only had half the time of the fieldbus technologies to get broad manufacturer support, but two suppliers doesn't represent global acceptance. FDT/DTM uses these other fieldbuses for the underlying technology it needs to provide basic communications,

Despite these potential shortcomings, FTD/DTM is a valid part of the control system environment. It's just that it isn't well suited for control. Therefore the whole FDT/DTM, Foundation fieldbus, HART, Profibus story is like comparing apples and oranges. Each member of this list is a fruit, and each group has its proponent and detractors. However, each of these groups and associated softwares have completely different properties. So, until we show a healthy skepticism for any technology and do our homework, we engineers aren't fulfilling our duty. It's an engineer's responsibility to find the best tool for the task at hand, and make tradeoffs as needed to find the integrated solution that satisfies the most objectives without contravening any of the boundary conditions under which it must operate.

End users are now beginning to realize that it's the maintenance and diagnostic information capabilities of the device that will provide the lifecycle savings and justification for installing new devices. However, we must not succumb to the lure of the latest technology just because it's new. We must continue to critically evaluate all tools available to us, not just for ease of installation/initial use, but also for their complete lifecycle support. Remember, with today's leaner organizations, it's our duty as engineers to provide the information required by our leadership to make the best overall economic decision as well.

I may not be from Missouri, but I still want you to "show me," and so I'm asking those of you with successes or horror stories about using Foundation fieldbus, FDT/DTM or other industrial network technologies to send them to me to share with our readers. I also welcome your suggestions for other column ideas. Send them to me at Ian.Verhappen@ICE-Pros.com

Ian Verhappen, contributing editor
Ian.Verhappen@ICE-Pros.com

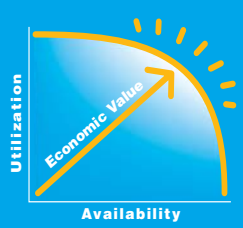


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Hannover Fair 2006 and the User Group Season

While industrial tradeshows in North America are dying, Hannover Fair in Germany continues to thrive. The Hannover Fairgrounds is the world's largest trade fair facility with 26 huge exhibit halls. Hannover Fair 2006, April 24-28, took up 17 of these exhibit halls, had more than 5,000 exhibitors from 66 countries, and attracted 200,000 attendees. Each of the 17 halls is immense, so visitors get to see the equivalent of an ISA Show, National Manufacturing Week, Promat, Sensors Expo, IMTS, Pack Expo, AM Expo and Quality Expo, all in one place, and all at the same time.

Vendor exhibits are also huge, perhaps two to three times the size of booths we're accustomed to seeing in the U.S. Siemens has a permanent exhibit that has 29,500 sq ft of floor space on the lower level, and 22,000 sq ft on the upper level, making it nearly the size of a football field. With such immense booths, exhibitors are able to display a huge number of products and some pretty big hardware. We saw an entire wind generator in one giant booth.

Halls were devoted to factory automation, power, industrial building automation, pipeline technology, subcontracting, industrial facility management and services, and the digital factory. Individual halls hosted their own forums, paper sessions, and events as sort of a "show within a show."

Interkama, Hannover's version of our ISA Show, took up six of the exhibit halls. Hall 13 hosted an industrial facility management and services forum with daily paper sessions,

FIGURE 1.

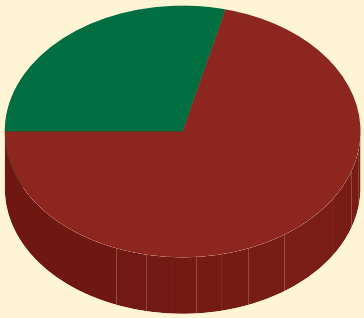
APC AT HANNOVER FAIR



American Power Conversion's exhibit at Hannover Fair shows some of the few signs in English at the event.

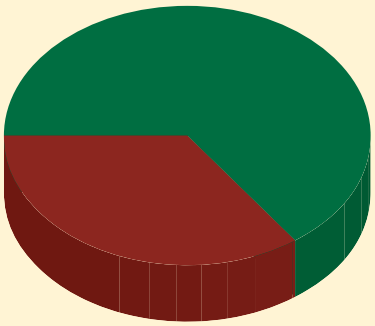
CONTROL Web Poll

Do you know there is a high-speed version of HART?



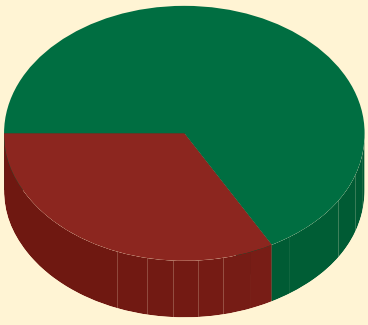
No 71% Yes 29%

If high-speed HART was available to you, would you do more with the digital signal?



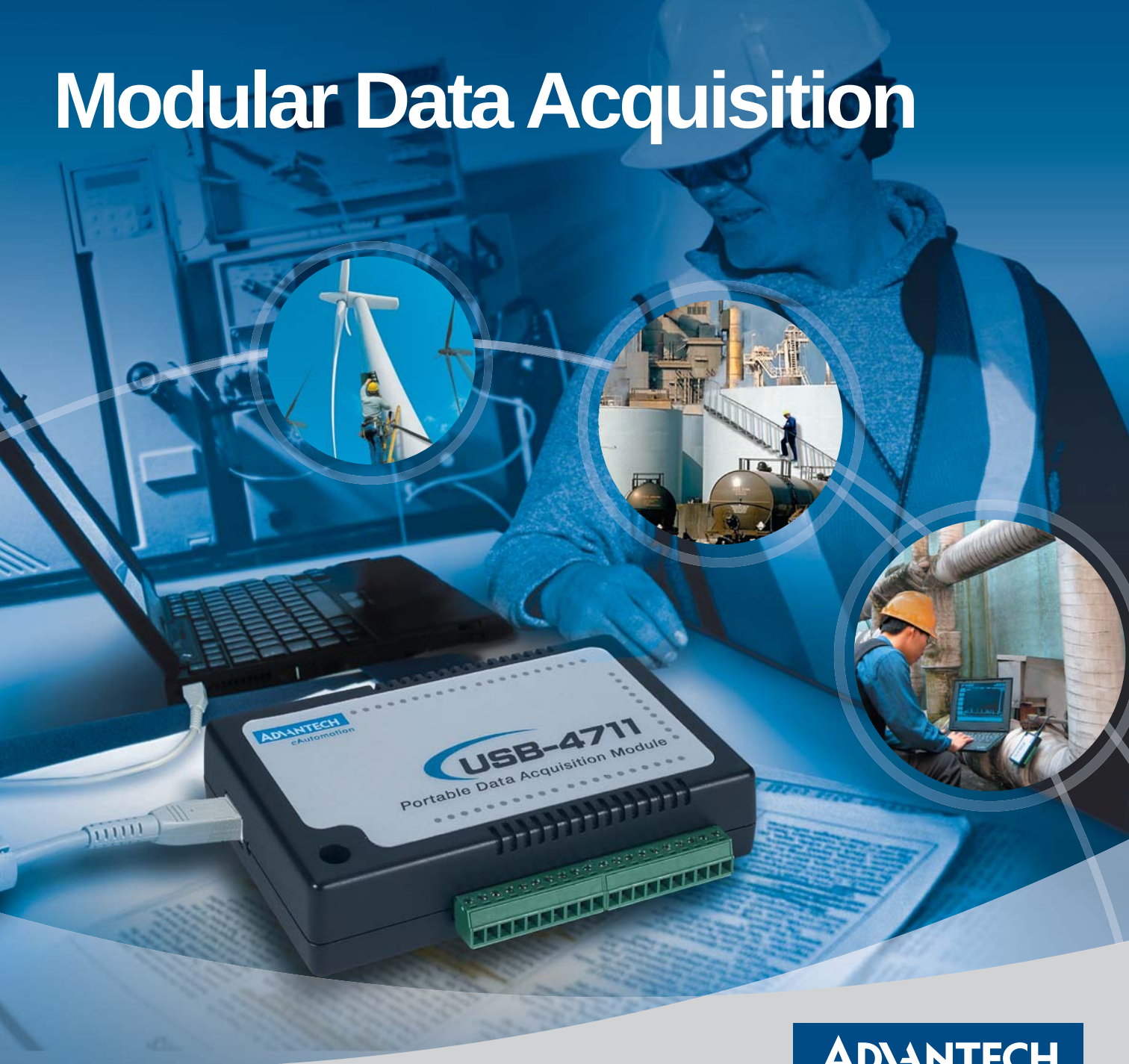
No 71% Yes 29%

Would you specify high-speed HART in your next project?



No 33% Yes 67%

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FIGURE 2.

HYDROGEN FUEL CELL



American Power Conversion's hydrogen-powered fuel cell fits in a 19-in. rack in the control room.

including speakers from vendors and industry. Hall 6 hosted a pipeline technology conference, and the World Energy Dialogue had paper sessions devoted to wind energy, fossil fuel power, power networks, oil and gas, and solar energy.

Clearly, you can find anything you want at Hannover Fair. Looking for new wireless technology? Forty companies exhibited wireless automation products. Looking for fuel cell technology? The Hydrogen & Fuel Cells group pavilion in Hall 13 had 100 exhibitors. You can see some amazing stuff, including hardware still in the Beta stage and advanced products, such as:

- American Power Conversion (APC) showed its new hydrogen-powered fuel cell that can replace diesel generators used for backup power. It's clean, neat, and fits in a 19-in. rack, right in the control room (Figure 2).
- Rittal displayed cooling units using CO_2 as a refrigerant, thermoelectric cooling using nanotechnology, and cold plate technology.
- Vietz showed a laser-based pipeline welding system.
- Harting introduced its new RFID transponder that has a range of 5 meters and can be used in the vicinity of metals and liquids, making it suitable for tough industrial applications.

Next year's show will be even bigger because it also will incorporate power transmission and control. But, if you do

go next year, brush up on your German. While nearly every exhibitor and attendee at Hannover speaks English, booth signs and literature handouts were mostly in German. So, if you don't know that "industriedienstleistungen und instandhaltung" means "industrial services and maintenance," or that "prozess-steuerung und Instrumentierung " is "process control and instrumentation," you could walk right by a booth that has what you're seeking. When you see familiar companies, such as Rockwell Automation or APC (Figure 1), you'll sigh in relief to see some signs in English.

In fact, its emphasis on German is probably Hannover's biggest problem, and may be why more North American engineers and press don't go. Last year, only 6,000 people came from North America, and only a handful of press. While its promoters tout Hannover Fair as an international show, it clearly is not. It's a huge German show aimed at a German audience. It just happens to have a large number of international exhibitors and attendees.

Still, it's worth a visit, especially if you're seeking new and innovative ways to solve plant engineering, maintenance, instrumentation and control problems. If it's new and innovative, it's at Hannover Fair. However, don't expect to cover it in one day, like you would a North American show. Allow an entire week.

Open Season for User Groups

Large vendors have responded to the decline of North American tradeshows by creating their own events, and using their own user groups as the vehicle. It could be argued that doing this has hastened the demise of the tradeshows in a sort of vicious circle. Still, there's no denying that between April 1 and Sept. 30, it's User Group season worldwide.



Easy as PI

The 25th anniversary OSIsoft User Group meetings were held in May in San Francisco and Frankfurt, Germany. Even though there are many people attending, the user group's gathering was intimate. One measure of the global nature of the OSIsoft User Group was the real-time translation service it provided, which was just like meetings at the United Nations.

The most interesting takeaway from the meeting was the many people who are working with the relatively new concept of a process plant (be it refinery, platform, power plant, or large enterprise in any process industry) as a collection of data, rather than concrete, tanks, pumps, field instruments, and control systems. Thus, the performance of the plant can be seen by drilling down into the OSIsoft's PI data server, and, as several speakers noted, very large sums of money can be saved.

As mentioned in this issue's Control Report column, "Is There a Tricorder in Your Future?" on p. 33, it's now possible to visualize data from the PI server at your plant through any Internet-enabled portable device, such as a cellphone, Blackberry, or other PDA. In an interview, Pat Kennedy, OSIsoft's president pointed to data streaming as the next important application for process-to-enterprise information management.



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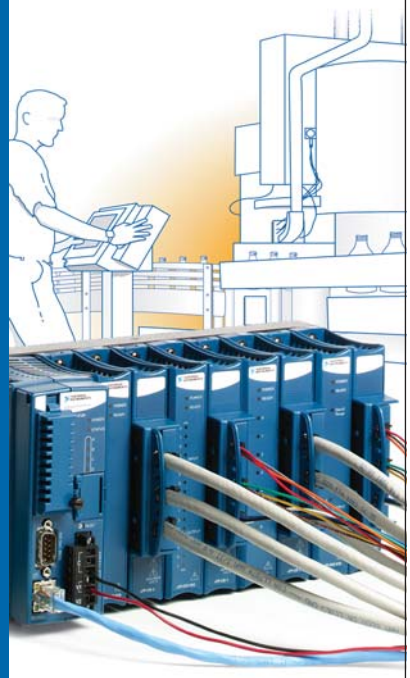
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ABB Blasts Off in Houston

Launched by an emotional lecture by former astronaut Dr. Jerry Linenger, one of the heroes of the MIR space station fire in 1997, ABB Automation World was significantly larger than last year. Staged May 9-11 in Houston, this year's event attracted more than 2,600 visitors from all over the world (and nearly 1,000 more than in 2005), who attended presentations on using ABB's products. These included a terrific study presented by Dow on using System 800xA, as well as a little town in Florida that became a testbed for a new magnetic flowmeter design.

With 40,000 sq ft of exhibit space and 400 workshops, Automation World is growing to rival other major user groups meetings. While presenting the company's glowing financial data, ABB president Dinesh Paliwal pointed to reductions in liabilities, a profitable 2005 globally, and in North America, higher bond ratings from Wall Street and a major, continuing R&D effort as proof that "ABB is Back!"

ABB Automation World 2007 will be held on March 20-22 at the Caribe Royale resort in Orlando, Fla.

HIMA Dedicates New Effort in Americas

Longtime leader in the German process safety systems market, HIMA dedicated the headquarters for its new HIMA Americas (www.hima-americas.com) subsidiary on May 12, 2006, in Houston. HIMA Americas was incorporated in September 2005, and is led by president Sven Grone. Essentially a re-launch of a former joint-venture effort, HIMA Americas is a wholly owned subsidiary of it German parent company, HIMA, which was first established in 1908, and was a key creator of the process safety market with its invention in 1967 of the first safety relay.

The new company will focus on the U.S. as well as Canada and Latin America, and represents the latest move in the company's shift from "a German company acting locally, to an international company acting globally," says Steffen Philipp, HIMA's managing partner.

In fact, this new HIMA Americas effort caps an impressive growth streak for the still privately held and family-owned company, which has grown its revenues from $28 million in 2000 to more than $75 million in 2005. During this same period, the firm's revenue mix has shifted from an 80% reliance on the German market to more than 80% of its revenue now coming from outside Germany. Also, though its staffing has more than doubled to 450 employees, the company still reinvests 15% of its revenue back into research and development.

Thermo Electron, Fisher Scientific Merging

Thermo Electron Corp. and Fisher Scientific International Inc. recently agreed to merge in a tax-free, stock-for-stock exchange, which reportedly will create the leading provider of laboratory products and services in the life, laboratory, and health sciences industry. The new company will be named Thermo Fisher Scientific Inc., and is expected to have more than $9 billion in revenues in 2007.

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ISA Forms Two Wireless Groups

ISA's Wireless Systems for Automation standards committee (ISA-SP100, www.isa.org) recently agreed to form two new standards working groups—SP100.14 and SP100.11. The first, SP100.14, will define wireless connectivity standards optimized for the unique performance and cost needs of a wide range of industrial monitoring, logging and alerting applications. The second, SP100.11, will define wireless connectivity standards addressing a wide range of applications optimized, but not restricted to, the unique performance needs of control applications ranging from closed-loop regulatory control through open-loop manual control. Both groups will coordinate their activities to enable SP100 to provide a complete and integrated set of standards for industrial wireless applications.

The two SP100 working groups will define the upcoming wireless standard's OSI layer, security, and management specifications, including network and device configuration. They also will determine specifications for wireless workers, wireless first responders, and wireless automation networks operating in automation and control environments.

The two working groups will issue a request this summer for proposals on topics, including architecture, radio-channel model, network coexistence strategy, RF interference strategy, and security. SP100 also is seeking volunteers to help create it new standard. Interested participants should contact Lois Ferson at lferson@isa.org.

Fieldbus Foundation Cooperating with NAMUR Group

The Fieldbus Foundation (FF, www.fieldbus.org) recently formed a liaison relationship with Working Group 2.6 "Fieldbus" of NAMUR, the Germany-based, end user group for the international process industries. Their cooperation will focus on "grounding and shielding" and "device diagnostics profiles." Both organizations identified these two topics as areas requiring further clarification and guidance, which hey say will benefit process industry users across the Europe, Middle East, Africa (EMEA) region.

Led by Stephen Mitschke, FF's fieldbus products manager, a working group has been established to investigate the existing portfolio of self-check and diagnostics for field devices. This initiative will ensure that Foundation fieldbus devices are consistent with the NAMUR working group's NE107 guideline that requires fieldbus devices to deliver extensive diagnostics to ensure that optimum plant efficiencies can be achieved.

Mitschke also is leading a second collaborative working group, which is compiling a guide to help end users make informed decisions about best practices for grounding and shielding fieldbus networks. FF's new guide is striving to comply with the NAMUR group's under-development NA114 guideline and with "international practice and options for grounding and shielding of fieldbus systems."

Controls Indices Show Robust Market

The National Electrical Manufacturers Association's (NEMA, www.nama.org) Primary Industrial Controls Index increased 6.1% between 2005's fourth quarter (4Q05) and 2006's first quarter (1Q06). This marks the index's 10th consecutive quarter of year-over-year growth, and constitutes its highest reading since the first quarter of 2001.

At the same time, NEMA's Primary Industrial Controls and Adjustable Speed Drive Index reached a new all-time high for its five-year history. This broader index, which includes adjustable speed drives, posted a 5.8% quarter-to-quarter growth rate during 1Q06.

"While part of this growth reflects typical seasonal demand growth, market conditions for automation and controls clearly remain strong because sales jumped more than 12% compared to the same period a year ago," says Brian Lego, NEMA's economic analysis director,

NEMA reports the near-term outlook for U.S. manufacturing bodes well for industrial controls and adjustable speed drives. Capacity use rates continue to edge higher, and output from non-high-tech manufacturing remains robust. In particular, business investment likely will push manufacturing activity even higher during the rest of 2006.

Boyes Receives Distinguished Service Recognition from ISA

Control's editor-in-chief, Walt Boyes, will receive ISA's Distinguished Society Service Award at the Honors and Awards Banquet at this fall's ISA Expo in Houston. Boyes has spent more than 20 years as an active volunteer with ISA in a variety of positions, and is presently chair of the upcoming ISA Management Division's Marketing and Sales Conference on Sept. 13-15 in Austin, Tex.

Is There a Tricorder in Your Near Future?

A zillion years ago, when I worked for NASA, we used an IBM 7094 Stretch computer to process ballistic camera data and track missiles being launched from Cape Canaveral. The computer room was nearly the size of a football field, with cabinets, tape drives, and core memories stretching into the distance.

The 7094 Stretch was the king of computers in its day, the forerunner to today's IBM BlueGene L and Cray supercomputers. It staggers the mind to realize that even a moderately powerful handheld PDA or cellphone has more processing power and 1,000 times more memory than the 7094 Stretch. These handhelds are becomng increasingly powerful, and will change the future of process control and maintenance. Soon, you'll all be carrying one.



Running the handheld as a thin client means the bloatware runs in a PC, but the handheld can see all the screens.

My fellow columnist, Dan Hebert, thinks so, too. His column, "The Right Tool for the Job," Technically Speaking, *Control*, March '06, p. 88, reported that HMI applications are coming to PDAs and cell phones. At the recent National Manufacturing Week 2006 show in Chicago, I found that maintenance people already like handhelds because they're easier to work with than the tiny, two-line LED display and four pushbuttons on most instrumentation. With wireless connections, they can calibrate and troubleshoot a device without having to crawl into a cabinet, or freeze their butts off on top of a distillation column in the winter.

Techs and supervisors who have to make inspection rounds love handhelds. No more writing on clipboards. No more entering work orders into a PC at end of shift. Maintenance and repair work orders can be entered, updated and reviewed in the field. No more paper at all.

Control engineers and operators are learning that they can go out into the plant carrying a wireless handheld, and see the same HMI displays that are on the control room's terminals.

Problem is, few vendors have taken handhelds beyond simple applications. Vendors program handhelds to do one task, usually for their own devices.

If someone combined HART and fieldbus communicator software with HMI/SCADA, vibration analyzer, device diagnostics, and mobile CMMS software, you could do everything in one handheld. It would be Mr. Spock's Tricorder, 300 years ahead of schedule.

One reason vendors aren't already doing this is because few truly understand the handheld's capabilities, and don't know how to make their software work diskless. Running software in a diskless environment is a deal-breaker for many companies.

Still, the Tricorder may not be far off. Handheld technology is growing by leaps and bounds. Already, you can get handhelds with 2 GB of memory, 300 MHz processors, GB plug-in memory cards, and wireless web interfaces. This power allows nearly all your Windows-based programs to run in a handheld, provided the programs can operate in a diskless environment.

It's cheap, too. A 128-MB cell phone can hold an HMI/SCADA system (3 MB from Indusoft), CMMS mobile application (48 MB from MRO), and Foundation fieldbus communicator (32 MB from Emerson Process) software, all at the same time, with plenty of room left over. Whatever you can't fit in the handheld can run as a "thin client."

Fabio Terezinho, product manager at Indusoft (www.indusoft.com), says not only is Mr. Spock's Tricorder possible, but it could be produced easily. "You can install and run application software on the handheld or use it as a Web thin client," he explains. "Using a database solution based on ADO (ActiveX Database Object) over TCP/IP, any mobile device can write/read data from any SQL relational database, such as SQL Server, Oracle, or Sybase."

Running the handheld as a thin client means the bloatware runs in a PC, but the handheld can see all the screens. From the handheld, you can access data from historians, ERP, MES, and similar databases.

Michael Saucier, of Transpara (www.transpara.com), agrees. His company showed software at the recent OSIsoft User Group meeting that permits bidirectional drilldown access to a company's PI server from any internet enabled mobile device. This includes your cellphone, PDA, or Blackberry, and it knows what device you're using.

It will probably take a small company on the bleeding edge to bring you the Universal Process Control/Star Trek Tricorder, but it's right around the corner.

***Rich Merritt,* senior technical editor**
rmerritt@putman.net

All Around Loop Controllers

***Control's* Monthly Resource Guide**

Every month, *Control*'s editors take a specific product area, collect all the latest, significant tools we can find, and present them here to make your job easier. If you know of any tools and resources we didn't include, send them to wboyes@putman.net, and we'll add them to the website.

TUNING GUIDES

How-to guides cover digital tuning, deadtime compensation, coarse/fine control, ratio control, compressor control, and other topics. Each describes the procedures and mathematics involved in tuning modern, digital controllers. Some are specific to the company's products, while others just use the products as an example of how to tune. To view these documents, go to www.sea.siemens.com/ia/support/inPIApps.html, and click on "Procidia Control Solutions - 353, 353R, 353P." **Siemens Energy & Automation; 215-646-7400 x2592; www.sea.siemens.com/ia**

ONLINE CONTROLLER SELECTION TOOL

This tool enables an engineer to enter temperature-loop controller requirements in the order of importance based on the application. Radio buttons and drop-down menus allow selection of various criteria such as thermocouple input type, control output, alarms, approval bodies, size, etc. The user is then presented with specific controllers that meet these criteria. Models can be compared for key attributes. To use the tool, go to http://catalog.sensing.honeywell.com/imc/ss.asp?FAM=controllers. **Honeywell Process Solutions 800-223-8947; www.honeywell.com**

CONTROLLING PH WITH PID

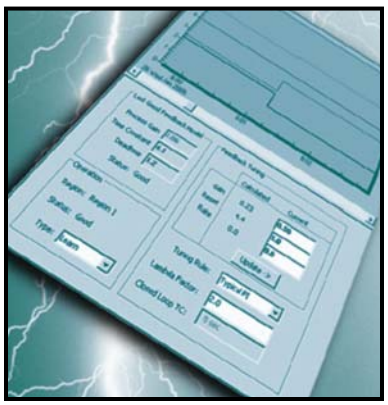
An application note on AN-320, "Overcoming the Challenges of pH Control," describes problems such as nonlinearity, long process deadtimes, and various disturbances that make controlling pH difficult. The note explains two common approaches that help PID controllers handle pH applications—linearization and gain scheduling. To obtain a copy, send an e-mail to controlsoft@respond1.net. **Control Soft; 440-443-3900; www.controlsoftinc.com**

PROCESS CONTROL TRAINING CLASSES

Courses teach PID, cascade, feedforward, gain scheduling, and multiple PID override, as well as override control, long deadtime processes, and interacting-loops theory and applications. Courses will be held in Tacoma/Seattle (August), Cleveland (September), Kansas City (October), Memphis (November) and Tampa and Boston (December). **Control Soft; 440-443-3900; www.controlsoftinc.com**

CONTROL TUNING BOOK

Good Tuning: A Pocket Guide, 2nd Edition by Gregory McMillan summarizes all the practical considerations for tuning loops. It includes step-by-step descriptions of the three best, field-proven tuning procedures, a table of typical tuning settings, a summary of valve performance problems, logic diagrams for troubleshooting, and more than 70 useful "rules of thumb." This second edition includes a new section on methods for reducing valve backlash and stiction, and preventing excessive sensor lag and noise. New chapters include adaptive control and application recommendations, and considerations for various types of process equipment. **ISA; 919/990-9287; www.isa.org**



TUNING TIPS NEWSLETTER

"Loop Tips" newsletter and website offers discussions tips for loop devices and controller tuning. Subjects from recent newsletters include: "Do you always trust your flights to auto pilot?"; "Is something dead in your loop?"; and "Getting tired of trial and error tuning?" To view the tips, go to www.emersonprocess.com/education/looptips.asp. **Emerson Process Management; 800-338-8158; www.emersonprocess.com**

PROCESS CONTROL VIDEO TRAINING

Courses consist of multiple units, each consisting of a 25-40 minute videotape, two corresponding student guides, and tests that can earn ECUs. Courses in controller tuning topics include Process Control Basics; Understanding Control Loops; Loop Tuning and Valve Considerations; PID and Cascade Control Basics; Feedforward Control Basics; Process Terminology and Behavior; Manual, Feedback and Cascade Control; Feedforward Control; Ratio Control and Other Control Considerations. For more info, visit www.emersonprocess.com/education/catalogrev/plantweb/pprocess.asp. **Emerson Process Management; 800-338-8158; www.emersonprocess.com**

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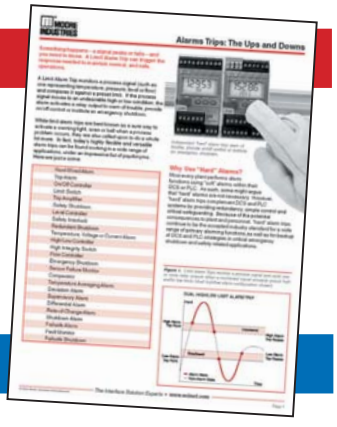
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Posting Gains

Respondents to *Control's* 2006 Salary Survey are collectively earning more money, increasing job security, reporting more hiring and fewer layoffs, and even growing slightly younger—on average.

by Jim Montague, executive editor

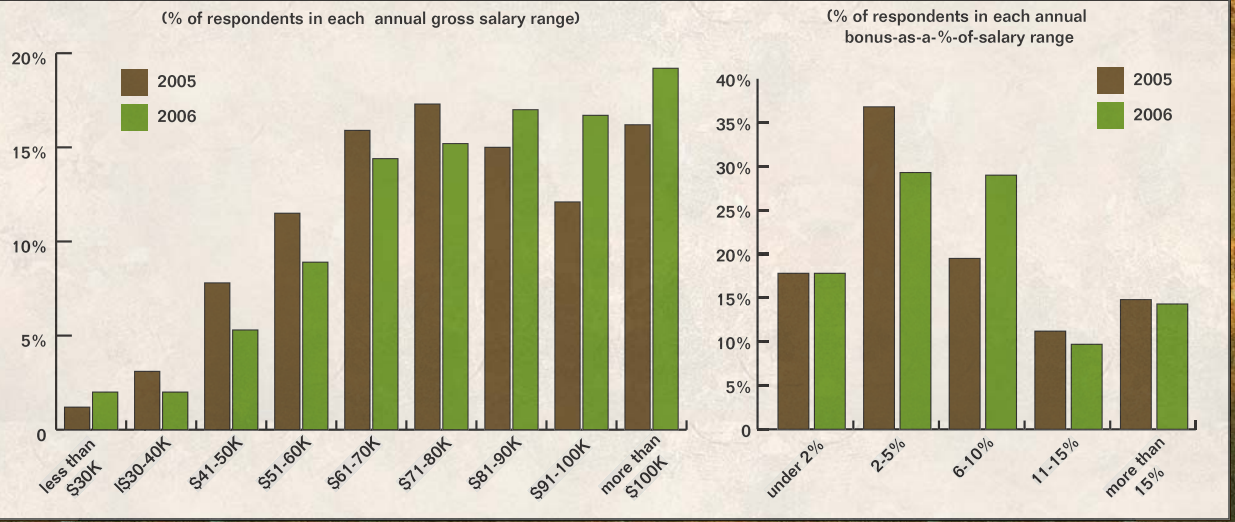
It had to happen eventually. Good news, that is. After years of receiving tiny, incremental pay raises, the process control engineers and other technical professionals answering *Control*'s 16th annual Salary Survey have finally made some substantial gains in compensation and other job-related areas. They're also less worried about job security, and report their companies are doing more hiring and making fewer layoffs.

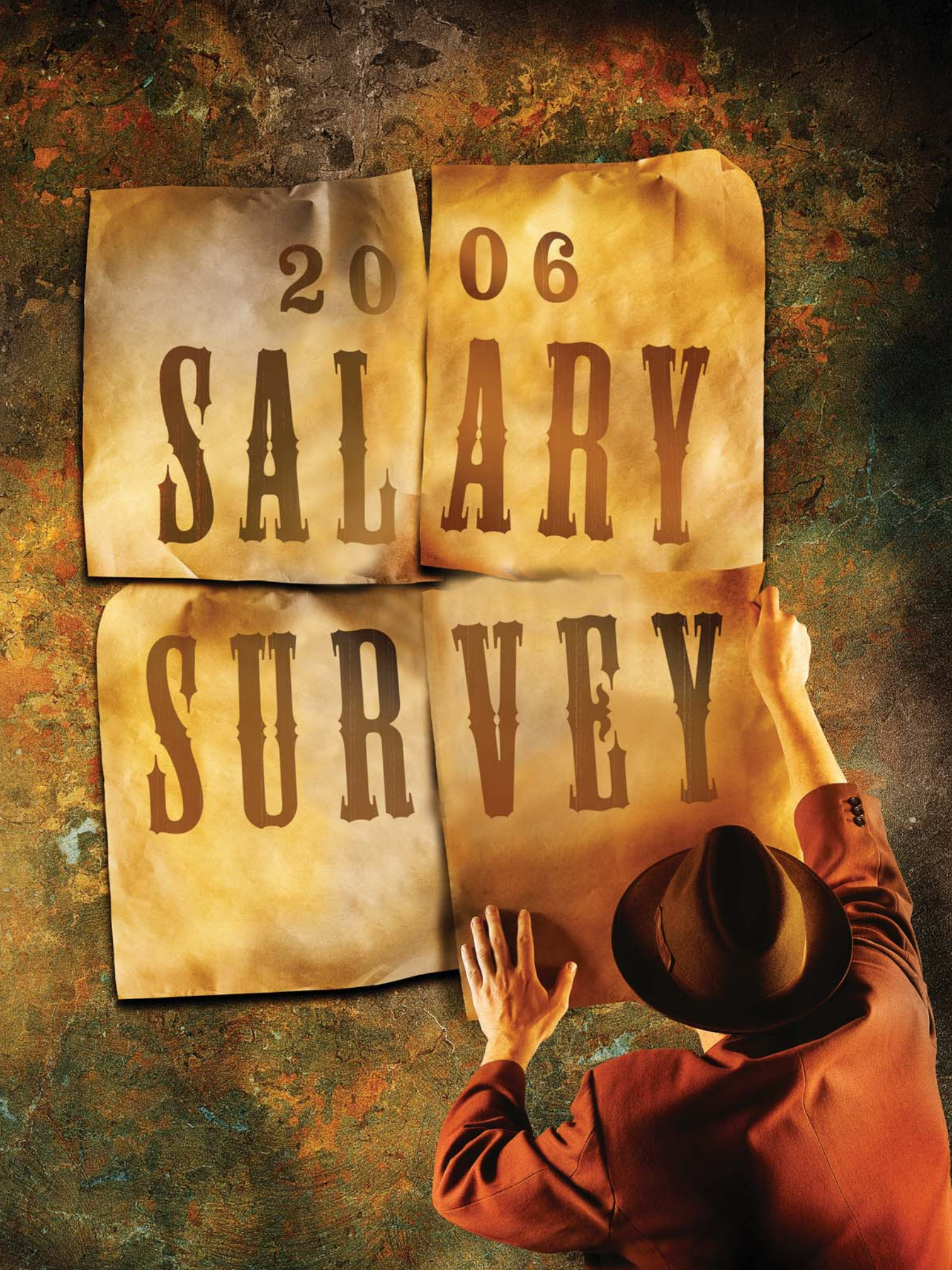
And the good news doesn't end there. In seeming contradiction to natural laws, the respondents have even gotten younger—at least as a collective group.

The survey's few negative results were that control engineers and their colleagues are working a couple more hours per week on average, and that their firms are continuing to increase outsourcing activities.

FIGURE 1.

HIGH SALARIES UP, SOME BONUSES BIGGER




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FIGURE 2.

TYPICAL PROCESS ENGINEER PROFILE

(by highest scoring % range in multiple survey categories)

	2005	2006
Salary	$71-80K (17.3%)	more than $100K (19.2%)
Hours per week	41-60 (78.9%)	41-60 (81.3%)
Bonus	yes (61.7%)	yes (63.4%)
Paid overtime	no (73.5%)	no (73.4%)
Vacation	three weeks (33.2%)	three weeks (32.4%)
Job description	engineering, design, construction (33.1%)	engineering, design, construction (38.1%)
Primary Industry	chemical (16.2%)	food and beverage (13.8%)
Years at present firm	11-20; more than 20 (24.1%)	11-20 (25.5%)
Number of jobs in career	two (25%)	two (27.2%)
Years in process control	11-20 (30.2%)	11-20 (34.8%)
Education	four-year undergraduate (48.9%)	four-year undergraduate (53.2%)
Degree	other, mostly mechanical engineering (36.3%)	electrical engineering (38.3%)
Residency	U.S. (98.3%)	U.S. (97.4%)
Gender	male (94.6%)	male (94.9%)
Ethnicity	white (86%)	white (88.9%)
Marital status	married (84.6%)	married (90.5%)
Children	yes (83.6%)	yes (84.1%)

Paydays and Profiles

The respondents to 2006's e-mailed salary survey reported a dramatic upward shift in their numbers in the two highest salary categories compared to the year before (Figure 1). Basically, respondents earning salaries above $80,000 increased significantly, while those below this threshold decreased slightly. Similarly, those receiving bonuses of 6-10% of their salaries increased from 19.5% in 2005 to 29.0% in 2006.

Besides improving their compensation, 2-4% more respondents in the 2006 survey reported receiving benefits than the year before in each category. These include medical and dental insurance coverage, life insurance, disability, pension and 401K plans, tuition reimbursement, stock options, profit sharing, company cars, flex time, and telecommuting. And, while three weeks of vacation remains the norm, those receiving four weeks increased from 24.4% in 2005 to 28.2% in 2006.

Demographically speaking, process control engineers remain overwhelmingly white, male, U.S. residents, who are married with children (Figure 2). Interestingly, a more than five-percentage-point jump occurred among respondents identifying themselves as married from 84.6% in 2005 to 90.5% in 2006. Another close to five-point increase was reported by respondents with four-year undergraduate degrees; 48.9% had four years of college in 2005, while 53.2% had four years in 2006. In addition, those possessing electrical engineering degrees increased from 34.4% in 2005 to 38.3% in 2006.

No doubt the most startling finding in the 2006 salary survey is that the respondents are getting collectively younger, though the majority remain middle-aged (Figure 3). In fact, those aged 36-45 years increased to 36.6% in

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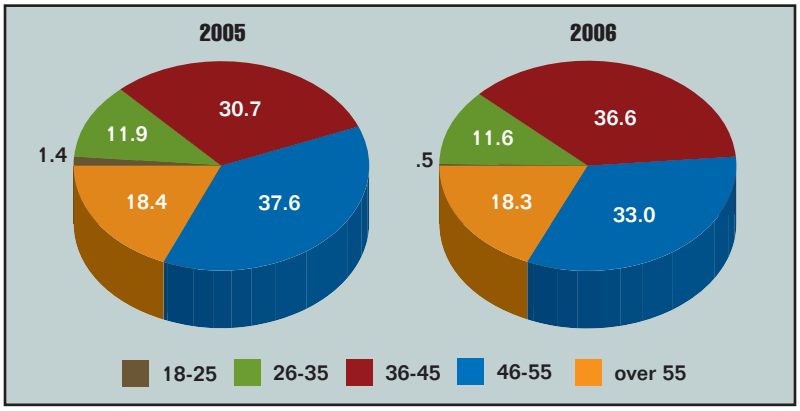
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FIGURE 3.

FOUNTAIN OF YOUTH?

(% of respondents in each age range)



2006 from 30.7% in 2005, while those aged 46-55 years decreased to 33.1% in 2006 from 37.6% in 2005. Respondents 26-35 years old and those over 55 years remained constant at just over 11% and 18%, respectively.

In addition, this youth movement occurred even though respondents with 11-20 years in process control increased to 34.8% in 2006 from 30.3% in 2005, and those in the "engineering, design, and construction" category increased to 38.1% in 2006 from 33.1% in 2005. As might be expected, however, respondents with two to five years of supervisory experience decreased to 21.8% in 2006 from 25.3% in 2005.

Larger Forces

The reason for many respondents' recent pay raises can be traced to several likely factors. Certainly, many of those in oil, gas, and other petrochemical-related applications are likely benefiting from recent financial gains in those industries. Likewise, increasing use of process control technologies in pharmaceutical and food and beverage applications is stoking demand for process engineers in those fields.

However, perhaps the most significant factor is the accelerating retirement of many older process control engineers, who aren't being replaced quickly enough. The usual lament is that too few young people are choosing process control and engineering in general as a career, but there are other forces at work too.

In recent years, control and automation engineers and other technical professionals moving through their careers have been challenged by a unique, overlapping, and shifting combination of technical, corporate, and economic influences. These changes have generated headaches and indigestion, at best, and disillusionment and despair, at worst, and too often make it harder to work and survive in process control and the many end-user industries it serves.

Besides the usual improvements in traditional technologies, this generation's control and automation engineers have coped with continually emerging software and increasingly PC-based solutions, multiplying fieldbuses and Ethernet implementations, plant-floor to enterprise connectivity, and wireless devices and networks. These galloping and often-overlapping technical advances have fueled frequent shifts and expansions in many engineers' job descriptions.

These events have, in turn, forced both veteran engineers and those just out of college to seek more continuing education to keep pace with new technologies. In short, one of the main reasons why there aren't enough process control engineers is because training is having a hard time keeping up

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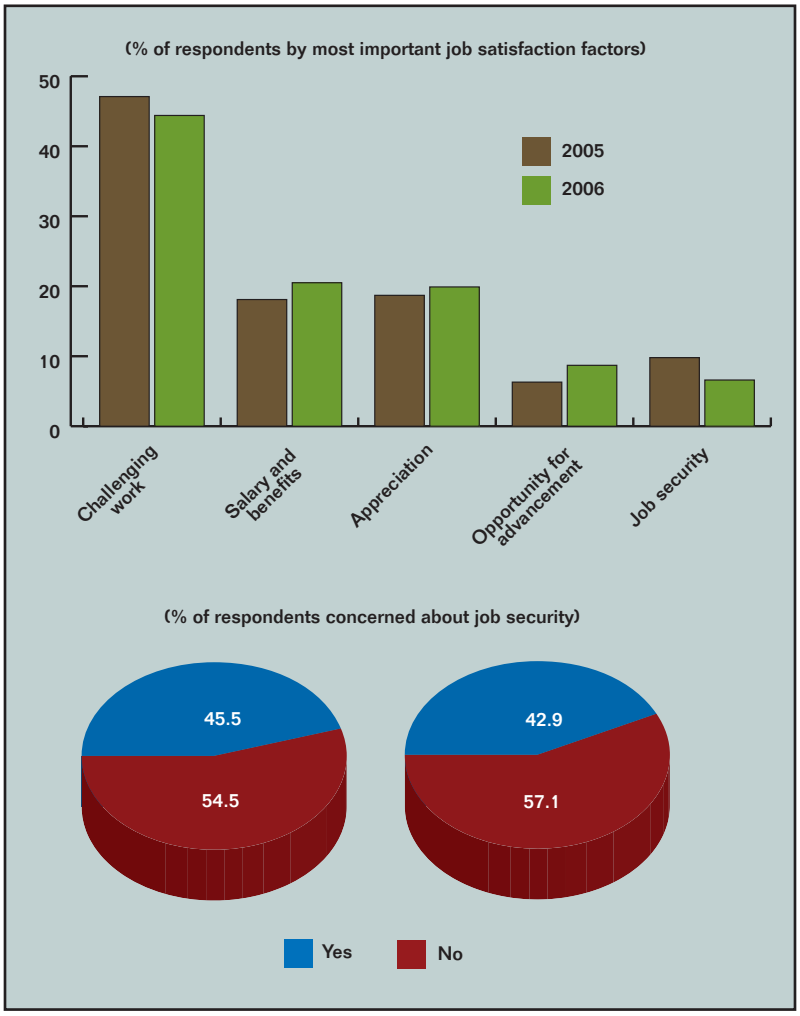
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FIGURE 4.

JOB SATISFACTION AND SECURITY



with accelerating technology changes. Bottom line: there may be increasing more financial incentive to train, but new technology and what you need to know keeps springing ahead anyway.

"I think we have about a five to 10-year gap in our engineering staff and their process control skills and experience," says Rob Stott, controls engineer in the APC Group at ConocoPhillips' LC Refinery in Westlake, La. "Most of us have 18 years of experience in process control, and are anywhere from seven to 17 years from retirement. We've always felt that we were somewhat behind in hiring, and so we made a concerted effort to recruit more control engineers, and then give new hires the two or three years of process control experience they need. However, we recently did a projection of the engineering staffing we're going to requires over the next 10 years, and we're going to have to do some even more significant recruiting and training to find the several dozen engineers we'll need. I don't know if we'll be able to find them."

Less Layoffs, More Outsourcing

The accelerating technology upheaval in recent years was compounded by a seemingly endless series of mergers and acquisitions, which were accompanied by rounds of layoffs that have only slowed recently. When Conoco and Phillips merged four years ago, for example, Stott says there was nat-

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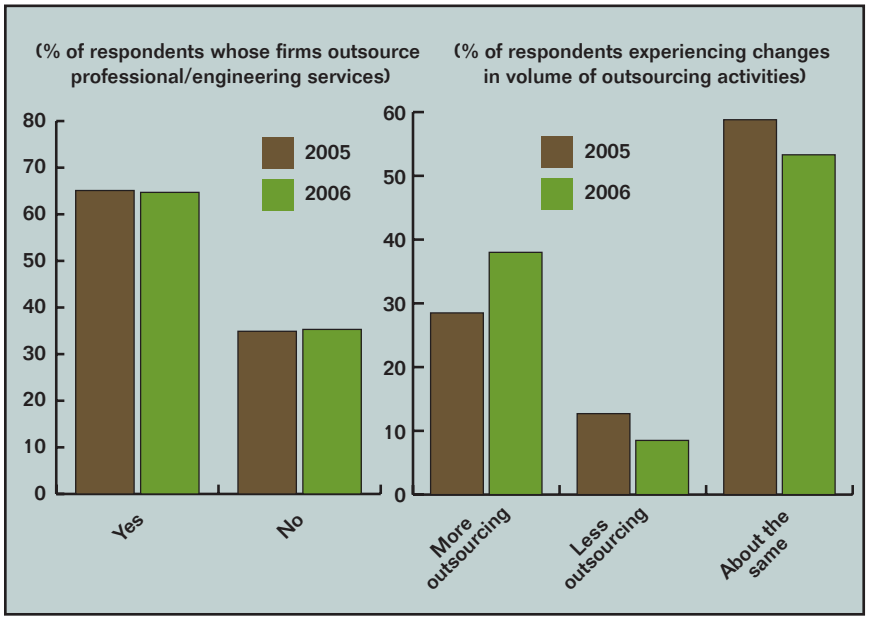
urally a lot of organizational shifting and some insecurity, but he believes things are calmer now. "We initially had a push away from centralized technical resources, but now there's some rebound back to it," says Stott. "Cost cutting now isn't focused on reducing headcounts. There's more concentration now on generating savings in operating funds."

The old merger-layoffs scenario apparently has been replaced by increasing efforts to outsource many engineering and other professional services. While the proportion of companies doing outsourcing remains constant at just over 65%, respondents reporting increased volumes of outsourcing shot up to 38% in 2006 from 28.5% in 2005 (Figure 5).

Conversely, some of the best news in 2006's salary survey may be the respondents reporting that hiring has increased

FIGURE 5.

ONGOING OUTSOURCING



CHOICE COMMENTS

Some respondents to the 2006 Salary Survey added highly enlightening comments to their usual multiple-choice answers. Several of the most interesting included:

- "The past two-plus years at our office have been brutal. A lot of people were laid off and many more were temporarily assigned to other offices for six to 18 months. Things are finally looking up, and we expect the next two-plus years to be good." — control systems engineer at California technical services provider
- "The engineering profession isn't one I'd recommend to a young person now. The level pay provides a decent income, but it isn't up to the effort required to get the degree and/or maintain the job. The turnover of executive leadership from engineering/manufacturing-based to financial-based has sapped most of the fun out of working in manufacturing in this country." — senior engineer at South Carolina paper manufacturer
- "The majority of our engineering and technical staff are between 40 and 60 years old. I don't see any young people in the workforce." — control engineering assistant at Texas electrical cooperative
- "We're running into issues with senior engineering personnel retiring. Much of their expertise hasn't been transferred to others in the group. We're dedicating resources to this area." — process engineering supervisor at Arizona petrochemical producer

continued on page 47



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at their companies from 38.8% in 2005 to 44.4% in 2006, while those reporting layoffs declined from 24.7% in 2005 to 20% in 2006 (Figure 6). These results were accompanied by an increase in overtime to 34% in 2006 from 29.9% in 2005. Unfortunately, promotions have declined to 11.8% in 2006 from 19.5% in 2005.

Meanwhile, those identifying job security as important for job satisfaction have dropped to 6.6% in 2006 from 9.8% in 2005. In addition, three-percentage points more respondents (57.1%) report they aren't worried about job security this year (Figure 4).

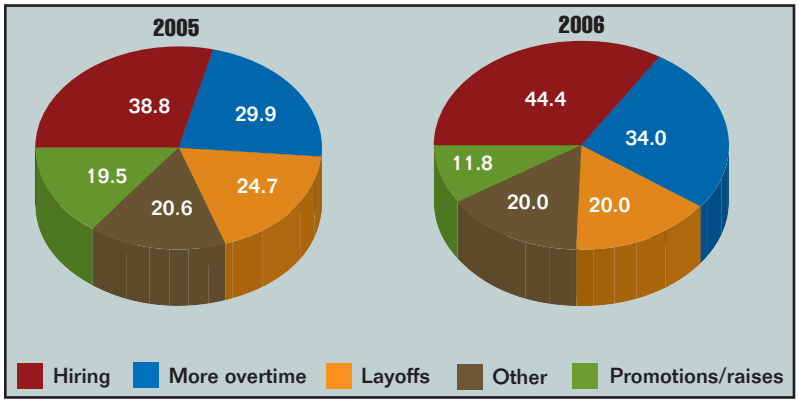
In fact, the increasingly hot market for process control engineers undoubtedly has some thinking that instead of waiting for a pay hike, they may "layoff" their employer, and get a better job altogether, despite the stresses involved.

"Many engineers probably think they're being underpaid by $15,000 per year, and that they could earn $30,000 extra by going to work for a systems integrator," says Del Ackels, controls specialist at Frito Lay North America in Vancouver, Wash. "However, they come to realize that they'd have to travel six months out of the year and deal with all the pressure of doing more startups, and suddenly they come back to the thought that they've got it pretty good. For example, I'm stretched pretty thin, but it's been that way since I began working as an engineer 32 years ago."

FIGURE 6.

ECONOMIC FACTORS

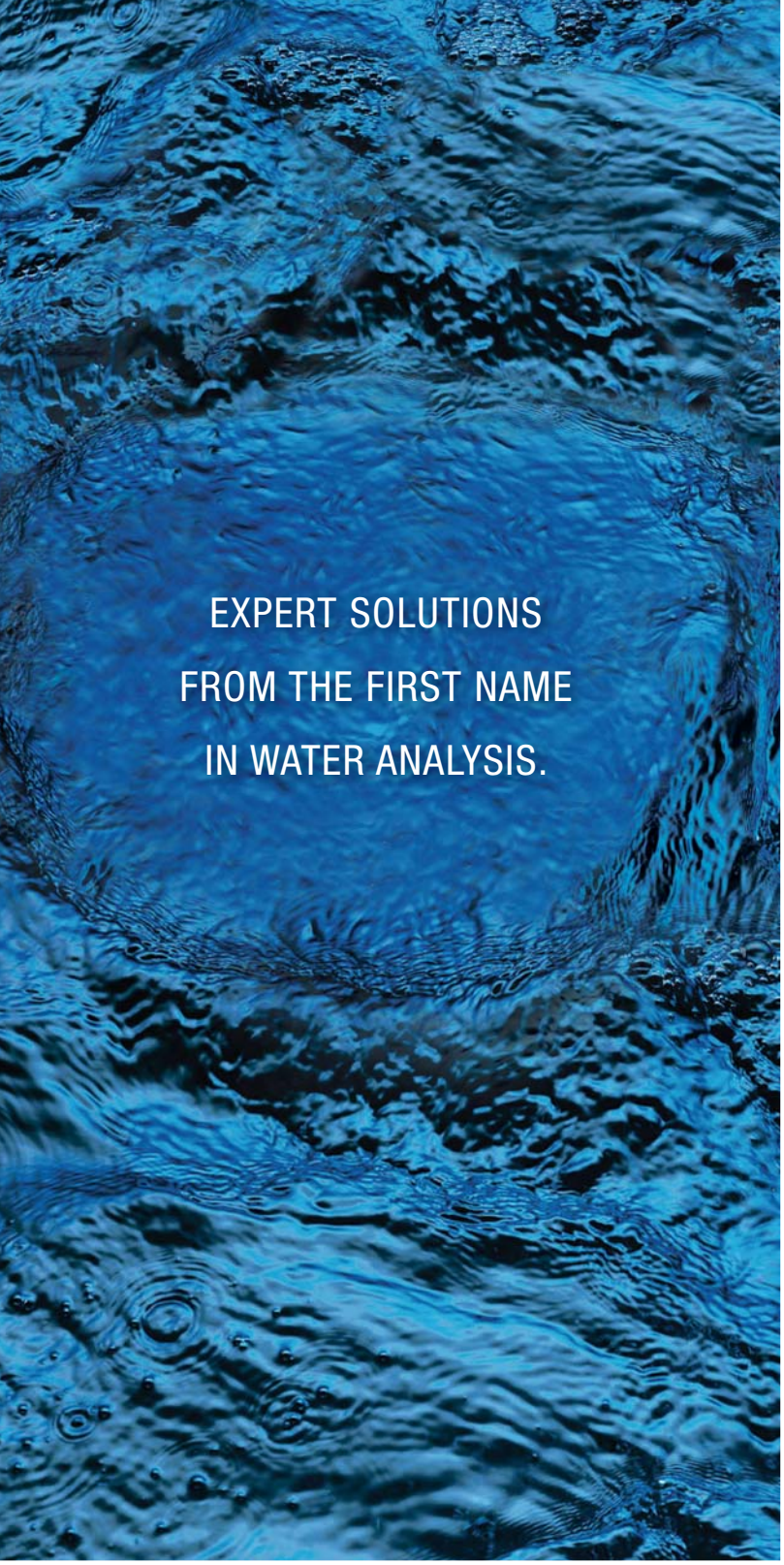
(% of respondents reporting influence of current economic factors)



continued from page 45

- **"Too many functions for everyone, and not enough time for core job responsibilities. I don't appreciate the trend away from people to automating everything." — senior instrumentation technician at Illinois oil and gas research facility**
- **"Overall employee attitude is horrible in these days when profit is the true #1 objective, while upper management says safety and quality are #1. Communication and honesty from upper management is nonexistent. Tell us the truth and the "whys" of some of the odd decisions being made. This is not the Land of Oz, so come out from behind the curtain. We can handle the truth and maybe even help resolve some issues." — senior process engineer at Wisconsin paper manufacturer**
- **"Our parent company is limiting equipment maintenance based on available dollars. This creates added maintenance for field operations because equipment that should be overhauled is being held together with bandages. Trying to explain these issues to directors is difficult because many don't understand operations or the impact of their decisions." — senior engineer at Texas natural gas producer**
- **"Outsourcing is used to to bridge the gap when we need another engineer, but just temporarily." — senior electrical designer at California engineering design firm**







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HOW TO LAUNCH AN ANALYZER SYSTEM RELIABILITY PROGRAM

Reliability is coming to refer to lifetime safety, quality, and costs as a means to safely and legitimately defer repair and replacement. **by Gary Nichols, PE, principal control systems engineer, Jacobs Engineering Group**

Analyzer Lifecycle Costing (LCC) is an accurate, descriptive, short name for an analyzer system reliability program, and "analysis and optimization of full lifecycle costs" is a functional, short description of the activities and results of a well-planned and well-run analyzer reliability program. This article describes and quantifies the lifecycle cost impact of typical analyzer system project issues and line items, which, if addressed early in the lifecycle of the project, can result in lifetime cost savings. It's important to note that "project," as used in the current context, means from concept to retirement, though retirement isn't addressed here.

One can't open a current issue of many trade publications without seeing an article with "reliability" in the title or section headings. Ricky Smith (*Plant Services*, Dec. '05, p. 54) notes that, surprisingly, reliability still may be struggling to become part of routine management. Related terms include "asset management" and "asset lifecycle management." Concurrent engineering (CCE) as Rich Merritt pointed out (*Control*, Oct. '04, p. 30) is a closely related concept and project management tool that involves "…multiple people, organizations and/or companies in a control system project," which "…goes beyond control systems to include the rest of the plant's operations and functions, from design and construction to startup."

Figure 1 shows the lifecycle cost of a hypothetical, but representative, analyzer system project. For simplification, we'll assume this is a small, standalone project to install a new analyzer system in an existing process unit. Figure 2 shows the cash flows for the same project. The costs don't consider interest, inflation, and tax consequences.

If we define the major lifecycle stages of a capital project or generic device as: 1) concept and planning, 2) design and development, 3) production and manufacturing, 4) operation and repair, and 5) wear out and disposal, then 65% of the project or device's total lifecycle costs have been determined at the end of concept and planning, and 85% of its total lifecycle costs have been determined at the end of design and development, according to Robert Dovich and Bill Wortman in *The Certified Reliability Engineer Primer, 3rd Edition*. Of course, this doesn't mean the money is spent and reflected as cash flow early in stages 1 and 2. Instead, its meaning is more subtle. For those with experience in capital project engineering and maintenance management, the implication is that ongoing maintenance, service, and lifecycle costs can be optimized (minimized!) with more thought and attention in the early lifecycle stages before capital equipment, direct labor, or direct materials have been purchased or, perhaps, even estimated. This article will use these and other assumptions, which may or may not apply to the readers' analyzer system projects and installations. Readers must work with their engineering, maintenance, and accounting organizations and relevant data to implement a valid, working analyzer system reliability program for a particular technical and business environment.

FIGURE 1.

LIFECYCLE COSTS

(1)	Capital cost of $150,000, assume capital completion during a single one-year accounting cycle
(2)	Engineering and design cost of 33% = $50,000 (expensed)
(3)	Custom order and spec items 33% = $50,000 (capitalized)
(4)	Direct labor and commodity materials 33% = $50,000 (capitalized)
(5)	Lifecycle = 10 years
(6)	Annual operating and maintenance cost of 10% of original project = $15,000
(7)	Dismantling and disposal cost negligible = 0

Capital project (expense and capital)	Year 1	$150,000
Annual operations and maintenance	Years 2-11	$150,000
Total lifecycle cost		**$300,000**

Production and Operation

Whether the figures of 65% and 85% apply to analyzers can be debated, but many readers can readily relate to stages 3 and 4 of a process analyzer's lifecycle, especially when unnecessary costs are incurred. For example, during stages 1 and 2, special DCS needs may have been overlooked; an improper sample-handling system alloy may have been specified; ergonomics may not be fully vetted; and basic principles of sample-handling system design may not have been fully considered. Any of these oversights can necessitate inconvenient and costly (safety, time, money, and

quality) reengineering, redesign, and replacement during commissioning and start-up. Even a Stage 1 decision can cascade to Stage 2 redesign because the design team failed to note that the analyzer is more critical during process plant upsets and other irregular operations than in steady-state or normal operations.

The following lists describe several typical project scope development and preliminary engineering-stage items for capital analyzer projects at stages 1 and 2. Though all should be addressed during these early stages, they're grouped to characterize the stage of an analyzer project when they're most likely to arise if not addressed early.

Project Scope Developments

- Analyzer scoped and specified on outdated process information has to be redone with current information
- Failure to specify correct sample handling system metallurgy
- Failure to recognize and request incidental infrastructure upgrades essential to the analyzer system during scope development
- International, national, industry, and company standards not clear about a non-customary feature on the analyzer system
- Failure to note need to purchase additional DCS hardware to handle multiple analyzer signals for stream component concentrations, safety-related alarms, functional faults, and maintenance actions
- Failure to recognize that no spare homerun wires exist between the field junction boxes and the motor control center or DCS marshalling panel
- Requests for post-completion documentation, job books, etc., not originally estimated nor approved for purchase or collation

Supplier-Related Events

- Competitive bids difficult to compare due to different proposed analytical technology, in-house and third-party capabilities, grouping of line items in proposals/bids, etc.
- Parallel analyzer system integration suppliers don't coordinate effectively
- Integrated analyzer system and associated purchased items don't pass factory acceptance test (FAT) the first time (and possibly not the second time)
- Supplier delays because of back ordering, internal reorganizations, personnel changes, etc.

Construction and Constructability

- Overlooking required safety features until construction begins
- Construction delays caused by operations or maintenance that could have been anticipated during scope development
- Awkward accessibility to construction site due to other work in the

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FIGURE 2.

CASH FLOW

Lifecycle Stage	Alternate Lifecycle Stage Names Typical for Capital-Engineering Projects	Cumulative % of Committed Cost	Committed Stage Cost	Cumulative Stage Cost
1. Concept and Planning	Scope development & preliminary engineering	65.0%	$195,000	$195,000
2. Design and Development	Detailed engineering	85.0%	$60,000	$255,000
3. Production & Manufacturing	Construction, commissioning & startup (8.3%*)	93.3%	$25,000	$280,000
4. Operations and Repair	Operations and maintenance (6.7%*)	100.0%	$20,000	$300,000
5. Wearout and Disposal	Dismantling and disposal	100.0%	$0	$300,000

* Assumed percentages of cost commitment in respective stage for discussion purpose.
Calculation Basis: Total Lifecycle Cost = $300,000

area, lifting near hazardous piping, lack of elevator or hoist for heavy capital items and construction tools
- Changing on-site construction contractors during construction
- Failure to note the need for special construction safety precautions during scope development and estimating
- Multiple personnel changes of engineers, designers, and other key project team members through project construction, start-up, and commissioning

These lists aren't exhaustive. The financial impact of each item in the above list is different for each project, company, and analyzer engineer, so it wouldn't be helpful to catego-

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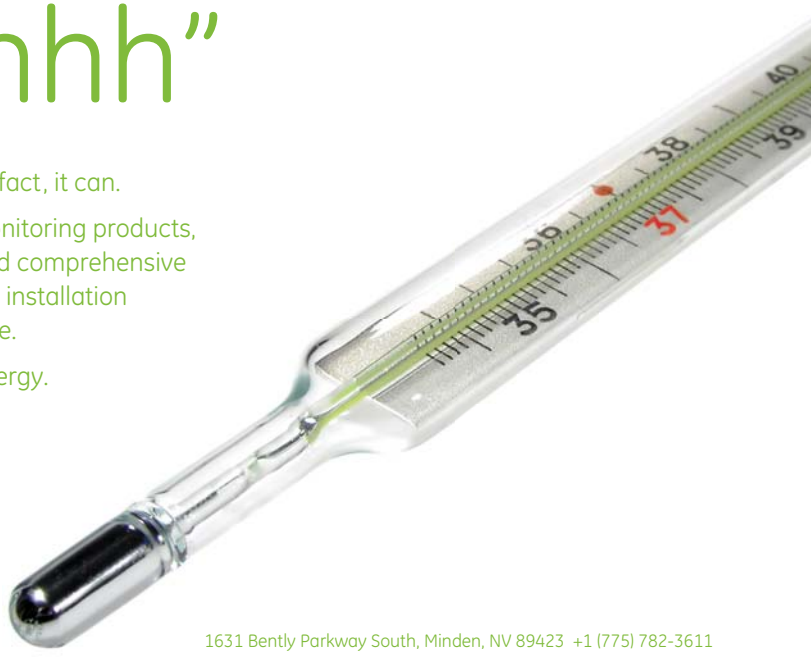
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	Actual Cash Flow by Stage	Actual Cumulative Cash Flow
	$25,000	$25,000
	$25,000	$50,000
	$100,000	$150,000
	$150,000	$300,000
	$0	$300,000

rize their respective impacts on safety, cost, schedule, and quality. If we look at the cost impact of a given analyzer project's problems by size, the important point is that, in each case—small, medium, or large, there's a considerable cost premium associated with discovering the problem and correcting it late during stages 3 and 4, instead of catching it early during stages 1 and 2. Just one each of the small, medium, and high impact items costs an additional $23,200 over the lifecycle of the project, and this doesn't include the costs associated with their unpredicted occurrence during Stage 4.

In addition, another $10,000 in capital costs can be depreciated at $1,000/year over a project's 10-year lifetime. This $10,000 can be justified if it will prevent a $100 per month, or $12,000 per lifecycle, service action over the same 10-year period. You can perform a similar analysis that includes interest, inflations, cash flow, tax consequences, etc. The point is that "a penny saved now is not necessarily a penny earned later."

The issue isn't ignoring nor minimizing the legitimacy of these concerns. Instead, the task is to know, understand, and consider their inclusion in the analyzer project scope as early as possible, and decide how best to include them, or to address them outside the scope of the project. Obviously, the correct metallurgy for an analyzer system must be in the scope, while completing incidental work not required for the analyzer system may often be addressed better and at lower cost outside the analyzer project's scope. Nor is it practical to avoid all such oversights. Sometimes a decision must be made for safety or financial reasons to change construction contractors. Sometimes it's necessary to work around unscheduled shutdowns and turnarounds for process safety or economic reasons. However, many oversights can be avoided during stages 1 and 2 if the right questions are asked of the right personnel at the right time. It's even better if the need for the information is recognized and volunteered early in the project lifecycle.

Historically, reliability was more associated with maintenance than with project engineering and management, as in the foregoing discussion. This notion is excusable because 1) an article or device that fails is judged to be unreliable, and requires repair or replacement, which are the customary functions of maintenance, and 2) a large body of practical input to improving the reliability of devices and systems comes from organizations that perform maintenance. More formal references to "reliability" often refer to its esoteric tools, such as mean time between failure (MTBF), mean time to repair (MTTR), design of experiments (DOE), redundancy, failure mode effects and criticality analysis (FMECA). These are only tools that can be used to measure the effectiveness of true reliability, which is lifecycle costing. Unfortunately, reliability is sometimes substituted for "maintenance" just to achieve a favorable image, rather than to make real improvements in the reliability of installations such as analyzer systems. Fortunately, reliability is coming to refer to lifetime safety, quality, and costs as a means to safely and legitimately defer repair and replacement, rather than as post-failure data collection and analysis, or as a meaningless, dressy name.

Gary Nichols, PE, is principal control systems engineer with Jacobs Engineering Group, and can be reached at gary.nichols@jacobs.com

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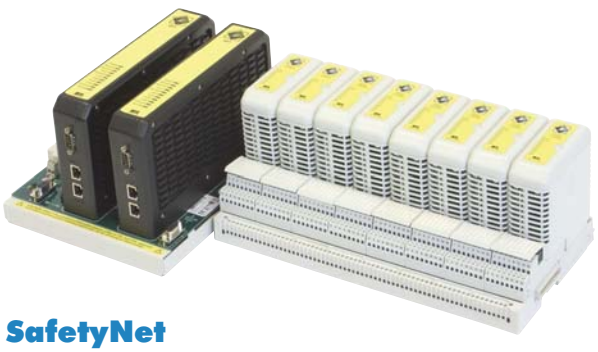
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THE ABC'S OF XML, PART 1

What you need to know to survive in the world of connected data

by John T. Sever, Cacade Controls

The project was nearly finished. The end was in sight after an incredible effort of long hours, weekends, battles, changes, and all of the normal challenges that accompany a large project. We'd delivered roughly 80% of the automation software for a life-sciences, S88 batching application when a problem surfaced–a big problem. Equipment modules and phases included commands to devices that could cross controller boundaries. By design, the code construct employed for these commands was non-deterministic when crossing controller boundaries. This meant that the destination may not receive the command. The problem appeared sporadically but rarely on our development system, and we could never recreate the issue in a controlled environment. On a large, plant-wide system with large I/O bus networks, the problem was suddenly pervasive.

The code had to be changed. To change one instance was simple enough, but we didn't know how many instances existed throughout the project's nearly 2,000 phases and equipment modules, each containing many steps that, in turn, contained multiple actions. My initial ballpark estimate to manually modify and validate the changes was between one and two man-years. Of course, the project didn't have money or time burning a hole in its proverbial pocket, nor were qualified resources available to do the work.

A few years earlier, this would have been a disaster for everyone involved. This was because we didn't know what was possible with XML (eXtensible Markup Language) and XSLT (eXtensible Stylesheet Language Transformation).

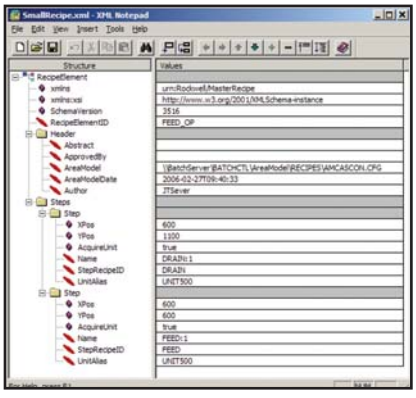
Once we understood the problem and the required code changes, we knew that only XSLT could help us make the necessary code modifications with minimal project impact. We created an XSLT transform to find every action with the faulty code construct, modify the code appropriately, and record each code modification in a change log file. The code was exported from the DCS system as structured text; converted to XML; transformed with XSLT to fix the problems; converted back to structured text; and imported back into the system. This XSLT transform was validated, so each change didn't have to be individually tested. All of the changes were completed with less than one man-month of work, no impact to the schedule, and minimal budget impact.

Despite these benefits, many in the world of automation haven't bothered to learn about XML or XSLT because they believe these technologies are for web developers, IT personnel, and consultants. The truth is that XML and XSLT are finding their way into nearly every aspect of computer technology today. In fact, XML-based technology has spread so pervasively and widely across application domains that many are unaware of its impact in their own discipline or of the potential value it brings.

This four-part series of articles will introduce the basics of XML, introduce XSLT as a query and transformation language for XML data, and provide examples of how these technologies can be used in automation. The three parts to follow are summarized below.

FIGURE 1.

XML READER



This tool takes the raw XML and translates it into an easy-to-read table.

Introduction to XML

XML is a flexible, general-purpose, data-storage structure in human-readable text format. Despite its name, XML isn't really a "language" like HTML (hypertext markup language). Instead, XML provides a set of rules for users to create their own markup languages. Anyone familiar with HTML will recognize many similarities with XML. These similarities made it easy for web developers to jump aboard the XML bandwagon. Its proliferation on the web has created some misconception that XML is a web-only technology. In fact Dictionary.com defines XML as: "*a meta-language written in SGML that allows one to design a markup language, used to allow easy interchange of documents on the World Wide Web.*"

However, XML is been used for much more than the easy interchange of documents on the web. So, a more appropriate definition might call XML: "*a meta-language written in SGML that allows one to design a markup language for easy interchange of data.*"

In this series, I'll introduce the basic rules of XML, terminology, and related standards. You'll learn how to read XML documents, and how to create your own XML documents. The series will also discuss declarations, processing instructions, namespaces, well-formedness, validity, DTDs, schemas, and encoding. Finally, I'll point you to a number of resources and applications to help you learn more about XML, and help you work with your own XML documents.

XML Revealed

Here's an example of what XML actually looks like, and as seen in an XML reader called XML Notepad (Figure 1):

```
<?xml version="1.0"
encoding="utf-8" ?>
- <RecipeElement xmlns="urn:
Rockwell/MasterRecipe" xmlns:
xsi="http://www.w3.org/2001/
XMLSchema-instance" Schema-
Version="3516">
<RecipeElementID>FEED_OP</
RecipeElementID>
- <Header>
<Abstract />
<ApprovedBy />
        <AreaModel>\\BatchServer\
        BATCHCTL\AreaModel\
        RECIPES\AMCASCON.
        CFG</AreaModel>
        <AreaModelDate>2006-02-
        27T09:40:33</AreaModel-
        Date>
<Author>JTSever</Author>
</Header>
- <Steps>
- <Step XPos="600" YPos="1100"
AcquireUnit="true">
<Name>DRAIN:1</Name>
<StepRecipeID>DRAIN</Ste-
pRecipeID>
        <UnitAlias>UNIT500</
        UnitAlias>
</Step>
- <Step XPos="600" YPos="600"
AcquireUnit="true">
<Name>FEED:1</Name>
<StepRecipeID>FEED</StepRe-
cipeID>
        <UnitAlias>UNIT500</
        UnitAlias>
</Step>
</Steps>
</RecipeElement>
```




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Introduction to XSLT

I recently overheard the following conversation between two automation engineers:

"They want to review the recipes."
"How do they want them?"
"Can you print them out?"
"That'll take a few days."
"Can you export them?"
"I can save them as XML."
"What do I do with that?"
"I don't know."

I'm not sure how this problem was resolved because I couldn't stick around

to help them out. If either knew about XSLT, they could have transformed those XML recipes into any number of formats. The easiest of these would have been a CSV that could be opened in Excel. Another option (my favorite) would be to transform the recipes into MS Visio drawings. Either of these two tasks could have been accomplished in a few short hours, and it would have worked on all 1,500 recipes.

XSLT is a standard for converting XML into something else. These somethings are growing rapidly. Most frequently, these are XML in a different data structure. This article opened with a problem, whose solution was to convert one XML file (with bad code) to another XML file (with good code). Other common formats created by XSLT include HTML and plain text. One of the most popular formats our company creates with XSLT is MS Word documents. Microsoft supports XML as a format for most or all of its Office 2003 applications.

In this series, I also will introduce XSLT as a declarative language. I will describe the transformation process and explain some of the basic XSLT programming elements. The series will also discuss XSLT expressions, match patterns, recursion, templates, default templates, XSLT processors, scripting, and some of the limitations of XSLT. Finally, I'll point you to a number of resources and applications to help you learn more about XSLT and to get up the XSLT learning curve.

Putting It All Together

Learning a new language is not a trivial undertaking. There needs to be a big payoff if you're going to spend the time to learn new data structures and languages. Rarely has a new language made your life better. This one will.

The final article will provide real-world examples of what can be accomplished with XML and XSLT. I'll make sure to point out obstacles that commonly arise and how to overcome these. I'll provide XML samples from real automation systems, and show accomplished tasks that have practical use in the everyday life of an automation engineer.

I hope that once we've finished this series you'll be well armed and ready apply these concepts to your everyday tasks.

Online Resources:
www.w3c.org
www.xml.com
www.xml.org
www.topxml.com
msdn.microsoft.com/xml

John T. Sever is president and founder of Cascade Controls, Inc., and has 20 years of experience as a process and automation engineer. He can be reached at 708/802-6000 or at johnsever@cascon.com.



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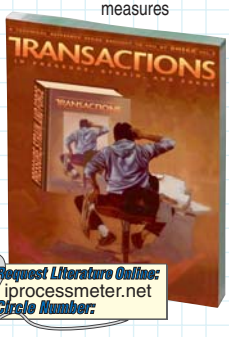
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INTRINSIC SAFETY VERSUS EXPLOSION PROOF

Why intrinsic safety never really caught on in North America

by Ian Verhappen, contributing editor

The purpose of Intrinsic Safety (IS) and Explosion-Proof systems is to prevent a malfunction in a piece of electrical process equipment from initiating an explosion or fire through ignition of gases that may be present in the surrounding atmosphere. Both systems do this by keeping the potential energy level below that necessary to start the ignition process.

IS manages the amount of energy available to a level below which ignition can occur. Explosion Proof contains the energy of any possible explosion within the enclosure containing the possible ignition source. This containment is done through careful design of the enclosure, so the resulting energy isn't only contained, but is dissipated through the large surface of the enclosure's flanges or threads. Consequently, if the integrity of the enclosure is compromised, either because of a scratch across the flange face or threads, or due to incomplete tightening of the cover, the result is a significant increase in the risk of an explosion. The net result is that Explosion Proof protection has a higher level of required maintenance than an Intrinsically Safe system (Figure 1).

Classifications

Class 1, Division 1 (Zone 0) areas are defined as having combustible mixtures present routinely or all the time. The probability of a combustible mixture in Div. 1 is defined as 1 to 10-1. This type of probability is a very high risk if arcing devices are used, so Div. 1 electrical design is very conservative. Div. 1 electrical designs allow for multiple simultaneous failures without the possibility of igniting a flammable mixture.

Generally, arcing devices for Div. 1 areas are sealed and contained in explosion-proof containers. A practical alternate to this approach is to limit power delivery to the extent that it is impossible to generate a spark. This alternative is Intrinsic Safety.

Class 1, Div. 2 (Zone 2) areas are defined as areas where the likelihood of a flammable mixture is low, but it's significant enough that normal electrical activity should not present an ignition source. The probability of a flammable mixture in Div. 2 is defined as less than 10-5. Single-failure tolerance with normal activities is much cheaper to design, construct, and maintain than the more redundant designs required for Div. 1. Div. 2 design is the most commonly found area classification in the U.S.

Unclassified areas have a negligible probability of hydrocarbons. This type of area classification might be found in an office building or control center.

FIGURE 1.

THE LIMITS OF POWER

Area Classification	IIC	IIB
Output Voltage (V DC)	30	42
Output Current (mA)	250	500
Output Power (W)	3.0	5.0

The approximate power limits for intrinsically safe apparatus in different area classifications.

"Intrinsic safety took off in Europe when transistorized equipment became available, and computer control became the norm. Possibly the principal factor for the different approach was the area classification."

Quick History Lesson

Chris Towle, secretary to several IEC standards committees and a long time practitioner in the field of electrical safety, provides the following background: "Intrinsic safety took off in Europe when transistorized equipment became available, and computer control became the norm," says Towle. "Possibly the principal factor for the different approach was the area classification. The existence of Zone 0, where the flameproof technique wasn't acceptable, required that IS had to be used for some sensors, and since it had to be used for this purpose, it became a recognized practice. There was no corresponding requirement imposed by divisional classification. In 1960-80, most of the investment in Europe was in petrochemicals and pharmaceuticals. This was predominantly done by companies with their own engineering capabilities that weren't involved in petroleum refining."

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FIGURE 2.

FIELDBUS SAFETY



Several devices are available for ensuring that Foundation fieldbus nodes are intrinsically safe. (Courtesy MTL)

More History: VDC on Protection Methods

The vast majority of users told Venture Development Corp. (www.vdc-corp.com) in 2002 that they used intrinsically safe and other protection methods such as explosion proof/flame proof, non-incendive and increased safety measures, and chose methods regarded as standard practice for their industry in their local markets. Intrinsic safety was perceived as a lower-cost solution when installation costs and lifecycle maintenance were factored in and compared with explosion proofing. Many users would consider switching to intrinsic safety equipment if there were significant savings.

In Europe, a larger percentage of intrinsically safe solutions was found for Zone 1 than Zone 2 regions. In North America, the largest percentage of intrinsic safety use was found to be for Division 2 regions. The chemical, oil and gas, and petrochemical industries are moving to Class 1, Div. 2 designations. There is evidence of users using reclassification, so there are more Division 2 and non-hazardous areas.

A small share of intrinsically safe I/O currently is implemented via intrinsically safe network/buses. Most of these are external products, with the rest internal to the local and distributed/remote I/O.

EPA Says 'No!'"

In the past, process leaks from leaking pump seals were very common, but were often tolerated as long as the leaks were small. Consequently, areas around pumps generally had a moderate likelihood of some hydrocarbon being present.

If these pumps were in an open area with free airflow and maintenance wasn't too lax, the envelope around the pump would be classified as Div. 2. If the pump was in an enclosed area, the same envelope would be classified as Div. 1.

The U.S. Environmental Protection Agency (EPA) made these pump leaks illegal in Div. 1 areas. We now have double-sealed or canned pumps, with monitoring programs to make sure ppm leaks are detected, corrected, and reported at far below the threshold of flammability. The likelihood of a flammable mixture has dropped a few orders of magnitude over the last 30 years.

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About the time that IS designs were made more widely known in North America, the U.S. started cleaning up leaks, much of the need for IS went away, and IS designs never caught on. Because of environmental changes, Division 2 designs appear quite robust, and Division 1 designs apparently aren't required in most installations.

Economic Factors

Joe Kaulfersch, market analyst for Pepperl+Fuchs Americas says, "Most users believe the cost is higher and that it's more complicated to have intrinsically safe circuitry. However, the cost of ownership actually is much lower, considering customers want to have higher system availability, be able to produce product without shutting down, and run the process safely. With explosion-proof equipment, the electronics must be locked out and turned off." Unfortunately, with fieldbus systems this isn't possible because they must be worked on live and connected to the network.

To help dispel these misunderstandings, MTL Instruments (www.mtl.com) has prepared a publication, TP1110-3, "Cost comparison of methods of explosion protection," which is available at its website. Based on MTL's analysis, the installed cost of an IS analog loop is 16% less than Explosion Proof, while for a discrete signal, IS is 17% less expensive.

FIGURE 3.

FISCO SAFETY



Intrinsically safe barrier, FISCO model (courtesy of SMAR)

ownership over a 10-year period by taking into account the effort required in permitting associated with maintenance, and found that IS was 38% less costly for a discrete signal and 34% cheaper for an analog signal.

The Fieldbus Effect

Fieldbus technologies will have an impact on the use of Explosion Proof and IS in North America, likely resulting in a reduction in the use of Explosion Proof, but not by replacing it with IS

When fieldbus technology was introduced to the market approximately 10 years ago, the only available options for power supplies were Explosion Proof and IS

"We use Non-Incendive rather than IS because IS simply doesn't make any logical sense if you follow it through."

Despite claims that IS installations allow "live maintenance," they and Explosion Proof require use of hot work permits in practice because many maintenance tools used aren't IS, and the work itself has the potential of creating a spark of sufficient energy to initiate a conflagration.

MTL's report investigated the potential impact of maintenance on

Since then, however, a number of alternatives have been developed, including several adopted by IEC and CENELEC such as FISCO (Fieldbus Intrinsically Safe Concept) and FNICO (Fieldbus NonIncendive Concept), which is only suitable for Zone 2 installations. Part of what

FIGURE 4.

ANALOG SAFETY



Typical analog intrinsic safety barrier, (courtesy of Cooper-Crouse-Hinds)

makes these alternatives possible is that Fieldbus is wired in parallel rather than series, so the energy is shared among the devices.

Several manufacturers also have developed other options that provide additional power to the field by reducing the energy level to the individual devices there. Among these options are the "High-Power Trunk" and "Split-Entity" concepts.

The High-Power Trunk, from Pepperl+Fuchs, sends up to 1A on the home-run cable to a field junction box in a Zone 2, where the energy is reduced to match the area classification, which could require an IS spur to the device.

As it name implies, Moore/Hawke's Split-Entity product "splits" the resistance of the barrier between the barrier in the safe area and the remainder of the resistance to the field spur to the device. The result is that more devices can be installed on one pair of wires.

Where We Are Today

Herman Storey, a senior internal automation consultant from a major oil and gas company, states "We use neither explosion-proof nor IS. Our plants are classified as Division 2, and neither EX or IS classification is required. I think plants in Europe will move to Zone 2 as well because of ATEX and the impossibility of complying with Zone 1 for mechanical equipment."

Storey adds, "We use Non-Incendive rather than IS because IS simply doesn't make any logical sense if you follow it through. You get into arbitrary safety factors that are compounded many times, and it gets expensive. Non-Incendive was designed with Zone 2 in mind, and it makes good risk management sense."

U.K.-based Chris Towle adds, "The only reason for sticking to Explosion Proof for instrumentation is inertia. I believe the major change in the use of IS will come with the acceptance of the 'ic' concept in Zone2/Div. 2 to replace the 'non-incendive/energy limited' concept. This should clarify the requirements for instrumentation in Zone 2, and lead to uniform practice applicable to all zones."

Dermot Coady, group business manager at MTL Instruments, adds, "One of the perceived limitations of IS is the unreliability of low-power solenoid valves. This was true in the early days of IS installations, but the design and reliability of these devices have improved so much that these former concerns are no longer an issue."

When asked why IS isn't being used in the U.S., Kaulfersch replies, "In North America, most hazardous locations are considered Class 1, Div. 2, which very closely resembles the Zone 2 specification in Europe. The fact is that many European factories are located very close to large populations. So, even though Zone 2 classifications are relatively safe, there never has been an accident on record with an IS installation. Insurance companies like to eliminate risks rather than mitigate risks. So, North American engineers are reluctant to use 'new' technology, even though it's 47 years old."

Storey summarizes the current state of affairs best. "I don't think the U.S. will ever move toward IS," he says. "It could go the other way, if anyone ever tried to make sense of the problem."

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"Collaborative Process Automation Drives Return on Assets," Dave Woll, ARC Advisory Group, June 2002

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Who Should Do Your Next Project?

If you choose to go outside your organization for assistance with your next process automation project, you have three main options: the automation department of a general engineering firm, an automation vendor, or an independent system integrator. Key factors determining your best option are your company's project specifics and internal capabilities.

In the area of specifics, you must determine project size, scope, and support requirements. Size is a key factor because contractors often assign personnel to projects based on size. If you have a $500,000 project, a $10-billion-per-year engineering firm won't assign their best people to your job. On the other hand, a $10-million-per-year firm is very likely to assign their top talent to your half-million-dollar job.



Your project should constitute no more than 20% and no less than 0.1% of the annual sales of your selected contractor.

Big engineering firms have outstanding depth and breadth of talent that's unmatched by smaller competitor, but execution of your project depends most on what people are assigned to it. In the end, companies don't execute projects, people do. Because the most critical factor affecting your project's execution will be the quality of the project team, you need to pick the right-sized firm to get the best team.

A good rule of thumb is that the size of your project should constitute no more than 20% and no less than 0.1% of the annual sales of your contractor. Projects above 20% can strain a contactor's resources, and projects below the 0.1% threshold may not get the attention they deserve.

Once the right-sized firm is determined, it's critical to pin down exactly who will work on your project. Again, the depth and breadth of a contractor's talent is not as important as the specific people assigned to your job.

Project scope is the next key factor in selecting a contractor. If your project scope encompasses only automation and system integration with one vendor's products, then an automation vendor or an independent system integrator with extensive expertise in the relevant vendor's products can be a good choice. "The main advantage of using a system integrator is in their expertise with a particular control system," says Richard Tullo, a project manager with Jacobs Engineering (www.jacobs.com).

Phil Murray, principal at system integrator FeedForward (www.feedforward.com), prefers automation vendors. "The control system vendor knows its platform better that anyone. If a vendor's instrumentation is provided in a digital fieldbus installation, it can stage the complete system."

If your project scope is limited to automation and system integration among products from multiple vendors, then a system integrator or an engineering firm can be of great assistance. However, if the project's scope encompasses multiple areas, such as facilities, process equipment, electrical distribution, and other general engineering areas, as well as automation, then a general engineering firm with a strong internal automation group may be your best option.

"Some clients like to work with one contractor for the entire project," says Bill Pollock, president of system integrator Optimation (www.optimationtech.com). "These clients contract the process design with the controls to an engineering firm such as Jacobs, so they don't have finger pointing."

Bruce Jensen, systems marketing and sales support manager at Yokogawa (www.us.yokogawa.com), adds, "An engineering firm also may be able to leverage systems integration pricing with other deliverables, and they can provide turnkey solutions at a lump-sum price."

Support requirements are the final project specific to be determined. Your company may need extensive or very little support, depending on internal capabilities related to the project's scope. Chief factors driving support quality are the caliber and proximity of the contractor's technical people.

If support is limited to product warranties, then an automation vendor scores high. Others can pass through a vendor warranty to some extent, but contracting directly with an automation vendor yields best warranty support.

If the only support needed is product warranties and minor and infrequent technical services, then the contractor's location is relatively unimportant. However, if extensive onsite technical services are required, then local support is critical. The selected contractor must have locally available personnel that can be quickly supplied at reasonable rates.

Next month's column will show how your company's own internal capabilities and characteristics drive selection of your best automation contractor.

Dan Hebert, PE, senior technical editor
dhebert@putman.net

Why is the Current Signal Standard 4-20 mA?

"Ask the Experts," on ControlGlobal.com, is moderated by noted process control authority Béla Lipták. Béla and his cadre of leading experts in process automation, recruited from among the co-authors of the *Instrument Engineer's Handbook 4th Edition*, are "in the box" all month, answering process control questions from all comers. The best will be published here.

Question:

My name is B. Shashikanth, and I have a master's degree in industrial process instrumentation. Presently, I'm working as a lecturer at one of India's engineering colleges. Even though I've completed my degree, I still have a small doubt in my mind. I discussed this question with my professors, but no one has the right answer. Sources on the Internet couldn't answer it either. My question is: why is the current signal standard 4-20 mA, rather than 3-19 mA or 2-18 mA?

***B. Shashikanth,* lecturer**

Answer:

Users Prefer 4-20 mA

After World War II, for nearly a half century, analog signals dominated our industrial control applications. Even the earliest analog signal ranges used a "live zero," so the failure of a transmitter would be distinguishable from a zero reading.

When the first edition of my *Instrument Engineer's Handbook* was published, pneumatic signal ranges of 2-14, 3-15, 3-18, 3-27 PSIG and 6-48 were in use, while 2-27 was used in power plants. In 1958, the SAMA standard RC2-1-1958 listed only three ranges: 3-15, 3-27, and 6-48 PSIG. As time passed, and due to the recommendation of user-dominated technical societies such as ISA, the 3-15 PSIG (0.2-1.0 bar) range became the international standard.

Some of the early electronic transducers depended on detecting resistance, capacitance, reluctance, reactance, and voltage. DC voltage ranges as low as 0-1 millivolts used potentiometer-type detectors. AC voltage wasn't used because of the difficulty of removing noise, inductive, and capacitive effects.

The early current signal ranges included 1-50, 4-20, 10-50 milliamperes (mA) DC. ISA accepted the 4-20 mDC range, which users preferred, and, after some hesitation by Foxboro, it became the *de facto* international standard. Two twisted-wire loops were used to run both the power and the signal to minimize common-mode noise and loading effects.

Power supplies were standardized at 24 VDC, with about 6 VDC needed to power the transmitter and 18 VDC remaining for wire and load resistance, IS barrier, and remote display drops, etc. For data acquisition and A/D converters, a 250-Ohm resistor was used to convert the 4-20 mDC to a 1-5 VDC signal.

When the digital age arrived, the technical societies became supplier-dominated, and international standardization became harder because each supplier wanted its own captive market, which resulted in the "fieldbus wars" and a "digital Babel." Though the ISO/ISA SP50 committee attempted to agree on an international fieldbus standard as early as 1980, it hasn't succeeded at the control level (Hunk 2, the data-link and application layers). In 2001, IEC published both Profibus and Foundation fieldbus, plus a variety of proprietory systems, as acceptable standards in its IEC 61158 standard.

While for reasons of both safety and economy it would be desirable that all digital "black boxes" use the same protocol (speak the same language), it will take some more time and a concentrated effort by users and their technical societies to achieve that goal.

Béla G. Lipták,* Control *columnist

Answer:

No Single Standardization Answer

Standardization is essential, which is why we have so many standards. The recent "fieldbus wars" were nothing new. The original IEC standard for electronic signal levels took more years to publish than it has pages! And, while it mandated 4-20 mA for two-wire systems, it also accepted 0-20 mA for three-wire systems with power on the third wire.

I recently worked on a power station system (circa 1968 vintage) that uses 0-10 mA field current transduced to 0-10 V through 1,000-ohm resistors. Then there was Honey-

well's Currentronik, which used a series 4-20 mA current through every module, rather than converting the current to voltage and tapping the users in parallel. This made it difficult to do on-line maintenance because any change killed every user.

In addition, Swartout used 1-5 VAC in early electronic instruments. Taylor used 4-20 mA through 62.5-ohm resistors to give 0.25 to 1.25 volts. Foxboro used 10-50 mA until the coming of intrinsic safety killed it. And, you can still buy 1-5 mV output process transmitters that will run for years on an on-board battery.

The live-zero came originally from the typical pneumatic instrument range (3-15 psi, 0.2-1.0 kg/cm2, 0.2-1.0 bar—all slightly different). While a transmitter will work down to zero-pressure output, it's hard to know whether you've got a negative error at the bottom end, and if linearity has suffered. These systems mostly used a common ratio, which is a quarter of the operating span for live zero. There was no particular logic that I can remember, but it carried over in the change to electronic signals on the basis that the technicians were used to the mathematics! So, there never has been one right answer. That is one of the joys of humanity.

Ian H. Gibson
consulting process, control, and safety engineer
(and rememberer of things past)

Answer:

History of Pneumatic-to-Electronic Signals

The early instruments were pneumatic, and due to their inherent construction, they couldn't work with zero output because the output signal was always a modulation of an air pressure. There was more than one signal available, but 3-15 psi undoubtedly was the most popular, and it still is in the electro-pneumatic transducers.

Other signals were used, like 6-30 psi for control valve driving, but the ratio between top and bottom value of the signal was always five. In Europe, the standardized signal was and still is 0,2-1 bar, which is very similar to 3-15 psi.

When moving from pneumatic to electronic, the ratio between top and bottom signals was kept in the large majority of the instruments, so we had 1-5 mA, 4-20 mA and 10-50 mA. This was true, at least, for the oil industry, which was a driving segment in the 1960s. The benefit of a live-zero signal is that, in case of problems in the transmitters or in the wiring leading to a zero signal, this abnormal condition can be immediately detected. A minor advantage is that when converting electronic signal to pneumatic, the conversion is simpler because the input and output signals are simply proportional, without the need to introduce bias. In the early days, this provided some advantage due to fewer physical components.

For some years, the three electric-signal standards (1-5 mA, 4-20 mA and 10-50 mA) coexisted, and I remember at least one instance in which controllers had 4-20 mA input and 1-5 mA output. Later, 1-5 mA was the first to disappear, while the other two remained until 10-50 mA, the standard promoted by Foxboro, was eliminated because its energy content was higher than allowed by intrinsically safe installations in dangerous areas. So, the conclusion is there isn't a philosophical reason why the signal can't be any one, and current situation is a step-by-step progress built on the past.

I hope I've explained the process by which the various signals originated and evolved, even though I haven't given you an answer explaining why 4-20 mA is correct and 3-19 mA is wrong. In my opinion, 3-19 mA isn't wrong, but it simply isn't supported by any international standard nor it is *de facto* standard like Modbus. In this age, when we all try to converge towards interchangeable and interoperable instruments, we have to reduce the options to the minimum amount of strongly justified differences.

Alberto Rohr
consultant and ISA Fellow

Answer:

4-20 mA Convenient for Contemporary Technology

The reason is simply because it's been found to be convenient for analog technology of the 20th century.

As you mention, a live-zero is handy for distinguishing between a "dead" connection/device and "zero" signal. It also allows powering a device from the signal. In addition, 16 mA provides a handy 4/1 signal range, and, with a relatively inexpensive (250-ohm, 0.25 W) resistor, can generate a voltage (1-5 V) that works well with analog readout devices of the same generation of technology.

Al Pawlowski, PE
Almont Engineering, Baton Rouge, La.

Pressure Instrumentation Growing Smarter

We are seeing the end of analog transmitters and proprietary protocols.

Sales of electronic pressure measurement devices are projected to increase at a 6% compound annual growth rate (CAGR) from $4 billion in 2005 to almost $5.4 billion in 2010, according to "Global Markets for Electronic Pressure Transmitters, Transducers, and Component-Level Sensors, 9th Edition" by Venture Development Corporation (www.vdc-corp.com).

This market consists of electronic pressure transmitters for industrial process applications, electronic pressure transducers, and electronic component-level pressure sensors. VDC defines electronic component-level pressure sensors as solid-state devices sold either as silicon chips or with limited packaging that allow them to be incorporated into pressure transmitters or transducers.

Nearly one-third of 2005's output of these products was for automotive-vehicle applications, which are increasing all the time. Applications include BAP (barometric absolute pressure), brake sensors, climate control sensors, diesel sensors, EGR (exhaust gas recycling), engine oil, fuel injection, fuel tank pressure sensors, MAP (manifold absolute pressure), power steering sensors, and tire pressure monitoring.

This is likely to be good news for the process control industry because the automotive industry buys in such large quantities that it drives sensor prices to rock bottom. By making pressure sensors dirt cheap, engineers can afford to put them in more places.

As for pressure devices in the process industries, VDC says "smart devices" are expected to increase to 91.4% of the market by 2010, with worldwide sales reaching $2.1 billion. This added increase is expected to come from displacement of traditional analog-output types, which is expected to decline from 10% of sales today to 7.6% in 2010. Could this signal the end of the traditional, analog, 4-20 mA transmitter?

A "smart" pressure transmitter, according to VDC, combines sensing technology and electronics in one package. The pressure sensor typically has a piezoresistive, capacitance, or other element that measures differential, gauge, or absolute pressure. The electronics typically provide one or more of the following features: remote calibration and ranging, self calibration, digital communications, digital output signal; self-diagnostics, pressure compensation, and temperature compensation. We can add wireless capability to that laundry list, too.

A very interesting section of the study concerns VDC's prediction of which communication protocols will grow, and which will fade. VDC says HART dominates in pressure instrumentation today, and is incorporated in 73% of transmitters. However, HART will likely fade to 67% by 2010, while Foundation fieldbus will double from 6% today to 13.9% by 2010, and Profibus and wireless Ethernet will emerge from zero today to 3.6% and 1.3%, respectively. Proprietary protocols from Yokogawa, Invensys/Foxboro and Honeywell are all predicted to drop market share, with Foxboro's Fox-Com nearly dropping off the scale at 0.8%.

Although VDC's prediction about communications protocols affects only pressure instrumentation, one suspects that it probably applies to other instrumentation as well. Welcome to the digital age.

Rich Merritt, senior technical editor

DIFFERENTIAL PRESSURE CONTROLLER

Series DH DigiHelic has 0.5%, full-scale accuracy on ranges from 0.25 in. w.c. to 100 in. w.c. Outputs include 4-20 mA, Modbus communications, and two SPDT relays with adjustable dead bands. Selectable options include security level; set point or set point and alarm operation; auto or manual alarm reset; and pressure, velocity or flow control. It can be powered by 120/220 VAC or 24 VDC. **Dwyer Instruments; 800/872-9141; www.dwyer-inst.com**



SANITARY PRESSURE TRANSMITTER

PI Series pressure transmitter is made from 316 stainless steel, rated IP69K, and withstands high-pressure cleanings and extreme washdown conditions. Accuracy is 0.2% full scale, and a four-digit integrated display provides continuous indication of pressure value. Application solutions include line pressure, filter monitoring, pump protection and tank level. More info at controlglobal.com. **IFM Efector; 800/441-8246; www.ifmefector.com.**



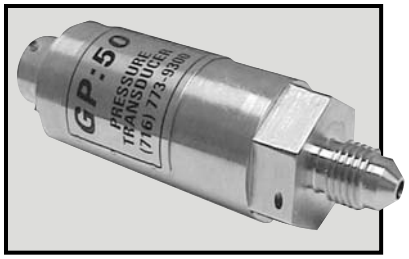
DIGITAL PRESSURE SENSOR

Sunx DP-100 pressure sensor has a dual display. The main screen displays current value and the sub-screen displays the threshold value or any four characters. Pressure ranges are from -14.7 to 145 psi with resolution of 0.0005 and a response time of 2.5 msec. Dimensions are 30x30x42.5 mm, and it can be panel or floor/ceiling mounted. **Panasonic Electric Works; 877/624-7872; www.us.pewg.panasonic.com**



PRESSURE TRANSMITTER

Series 42 industrial pressure transmitter has all welded, stainless-steel construction to withstand the rigors of automotive, industrial test or any application with high shock and vibration.



Outputs are VDC and 4-20mA output over ranges from 0-15,000 psi with accuracy to 0.1% FSO. Temperature compensation is -40 to 250 °F. **GP50; 877/774-4751; www.gp50.com**

PRESSURE TRANSMITTER

Barcon PPC has a ceramic sensing face for stability, resists corrosive and abrasive media, and is suitable for sanitary applications. It works with



gases, vapors, and liquids. Intrinsically safe models allow for installations in explosive areas with certifications to CSA, FM and ATEX standards. Options include fieldbus protocols such as Profibus PA or HART. **Pepperl+Fuchs; 330/486-0002; www.am.pepperl-fuchs.com.**

WIRELESS MONITOR

Unnamed wireless system measures and monitors pressure in Class I, Division 1 areas. It's available in battery powered, solar powered, and AC or DC powered versions. Applications

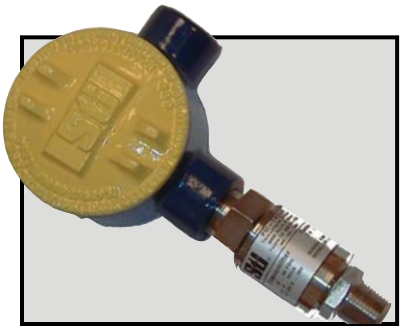


include tank farms, pipelines and general industrial pressure monitoring. **Adalet Wireless; 216/267-6864; www.adalet.com**



HAZARDOUS PRESSURE TRANSMITTERS

PT-L1 dynamically compensated pressure transmitters are cCSAus approved for use in hazardous locations. An intrinsically safe model is available to satisfy Class I, Div. I, Groups A, B, C



and D environments. An explosion-proof model is also available to satisfy Class I, Div. I, Groups C and D; Class II, Groups E, F, G; and Class III environments. More info at controlglobal.com. **Scientific Technologies; 888/349-7098; www.stiapg.com.**

PRESSURE CONTROLLER OPENS THE GATEWAY

Ethernet-enabled UT551 digital indicating controller has a Gateway



function that allows users to establish an Ethernet link between the pressure controller and DAQWorx software running on a PC or another host system. More info at controlglobal.com. **Yokogawa; 770/254-0400; www.yokogawa.com**

DIGITAL PRESSURE GAUGE BURSTS ON THE SCENE

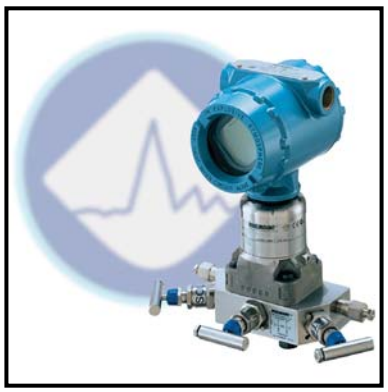
AccuGauge media-isolated digital pressure gauge works with any gas, liquid or solid media that is compat-



ible with 17-4 PH stainless-steel. It's suitable for applications that require high proof and burst pressures. Accuracy is 0.5% full scale including non-linearity, hysteresis, and non-repeatability over operating temperature ranges of -10 to 160 °F. More info at controlglobal.com. **AST; 973/448-1901; www.astsensors.com.**

SENSOR DIAGNOSES ITSELF

The scalable 3051S Series of pressure instruments has HART diagnostics and statistical process monitoring that provides early warning of abnormal process or equipment conditions. Examples include detecting plugged impulse lines and changes in wet gas conditions. Device operating hours are logged along with



the occurrence of diagnostic events, such as overpressure or overtemperature conditions. It has an enhanced EDDL graphical user interface. **Emerson Process Management; 800/999-9307; www.emeresonprocess.com**

WIRELESS TRANSMITTERS

XYR 5000 wireless transmitters enable automated monitoring of variables



in areas where traditional, hardwired transmitters are too costly, difficult or time-consuming to implement. Models include devices for monitoring gauge pressure, absolute pressure, differential pressure, temperature, and ultrasonic noise (for detecting steam and gas leaks). More info at controlglobal.com. **Honeywell Process Solutions; 215/641-3798; www.honeywell.com**

SEND IN THE MARINES

PX2471 Series pressure transducer meets the new Marine Pressure Standards MIL-T-24742(SH) and AS F2070. Applications include naval shipboard use, both on surface and sub-surface vessels, as well as general use on new construction and ship overhauls. It is directly interchangeable with existing pressure transducers designed to MIL-T-24742(SH). More info at controlglobal.com. **Omegadyne; 203/359-7749; www.omegadyne.com**

ALL-WELDED GAUGE

Solfrunt 1981 process gauge with



stainless-steel internals is mounted and welded onto a diaphragm seal. The diaphragm seal is available with either 316L stainless-steel or Hastelloy C wetted parts. The unit's all-welded seal-to-gauge design and seamless Bourdon tube provide fewer leak paths. Standard units have a female top for the instrument connection and a 1/2 NPT male bottom for the process connection. Other connections or materials are available. **Ametek U.S. Gauge; 215/355-6900; www.ametekusg.com**

COMPACT TRANSMITTERS



Model 2600T pressure transmitters are available in ranges from 1.2 in. H2O to 8,700 psig and 2.25 mm Hg to 435 psia with a 20:1 turndown and accuracy of 0.15%. The transmitters use a piezoresistive strain gauge or ceramic sensor, depending on the selected span. All models have a Hastelloy C276 diaphragm with silicone oil fill fluid. More info at controlglobal.com. **ABB Instrumentation; 800/922-2475; www.abb.us/instrumentation**

DIGITAL BAROMETER

The mercury-free Barocap PTB220 digital barometer has a silicon-based capacitive sensor which has a pressure measurement range of 500 to 1100 hPa and an operating temperature range of -40 to 60 °C. The backlit local display simultaneously shows barometric pressure, three-hour pressure trend, and the WMO pressure tendency code. All barometers are delivered with a NIST traceable certificate. **Vaisala; 783/981-4500; www.vaisala.com**



Multihole DP Orifice Flowmeter



Emerson's Rosemount 1595 Conditioning Orifice Plate is usable in high-pressure applications to ANSI 2500 and DIN PN100 and process applications requiring material compatibility such has hastelloy, monel and 304L SS. Rosemount 405C Compact Conditioning Plates offer a centering option for ANSI, DIN and JIS standards that facilitate alignment between pipe flanges, reducing installation time, and extending orifice plates into measurement points previously limited by piping configurations. **Emerson Process Management; 800/999-9307; www.emeresonprocess.com**

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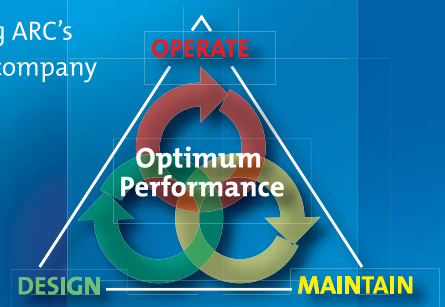
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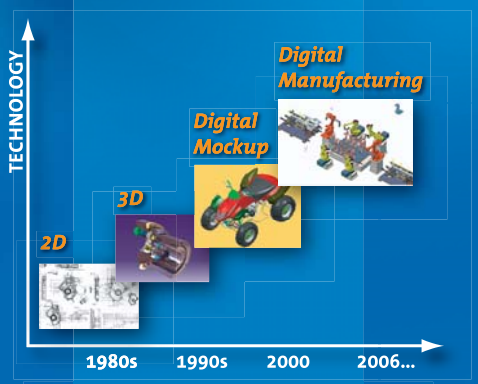
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Ringing Changes on Wet Sensors

Datastick product line enlarged with four new parameters

Based on the all-digital wet chemistry parametric sensor product schema they introduced last year, Aquasensors has released four new parameters for the Datastick family.

Datastick is based on a modular concept with interchangeable sensor heads in a variety of parameters, a universal sensor module (the "stick"), and interchangeable communications adaptors for a variety of digital communications protocols and fieldbuses. Already released parameters include pH, ORP, resistivity, conductivity, and toroidal conductivity.

Each sensor head is digital, pre-calibrated, and knows its serial number, sensor type, calibration data, and other diagnostics for validation purposes. Sensor heads are completely interchangeable without recalibration.

Sensor and communications power requirements are low, according to the manufacturer, typically 120 mW from a 10-30 V Class II DC power source.

Fieldbus and communications protocols available right out of the stick include Modbus, Profibus, DeviceNet, RS-485, RS-232 and USB 2.0 serial communications. According to Bruce Bathurst, president of Aquasensors, HART and Foundation fieldbus will be available soon. He notes that wireless sensor operation is available by taking the Modbus output into one of the commercially available wireless gateways, and thence to the control system. "This makes the system well-designed for remote monitoring sites," Bathurst says.

FIGURE 1.

ALL DIGITAL!



New DO, Ozone, Raw-Water and Drinking-Water turbidity sensors are fieldbus enabled straight from the sensor.

Local display and analog output are provided by a Modbus module, AV38, with display, programming, up to two current loops, and two Form C relays. Local display isn't required for operation of the all-digital system.

The four new parameters are: dissolved oxygen (DO), ozone, raw-water turbidity and drinking-water turbidity.

The DO and ozone sensors are similar because both are based on Clark Cell technology. They're supplied in a variety of body materials, two types of sensor tip protection, standard or custom filling solution, and either a FEP Teflon membrane or a custom solution.

Both the DO and ozone cells also measure temperature, and are air calibrated at the factory. The company claims accuracy of ±0.04 ppm with a resolution of 0.01 ppm, and a step response time of 90% in 90 sec over a 0-40 ppm range. Bathurst notes that these are typical at 25 °C. He also points out that performance of the sensor is unaffected by cable length.

The turbidity product is provided in two sensor types, raw-water and drinking-water turbidity. The raw-water sensor looks like all the other Datastick products, and like most other parametric sensors. It has a range of 0-4,000 NTU (Nephelometric Turbidity Units) with an accuracy of 0.2 NTU and a resolution of 0.1 NTU, with a step response time of 90% in 15 sec.

Similar to its drinking-water turbidity counterpart, the raw-water turbidity Datastick uses a quartz or sapphire window, with a multi-color source having detectors located at 90° and 180° to the source. Turbidity and temperature are pre-calibrated at the factory, and the sensor retains calibration and diagnostics for validation in food, beverage, and pharmaceutical applications.

In addition, the drinking-water turbidity sensor has a flow through cell into which the sensor inserts, and operates over a range of 0-100 NTU with a resolution of 0.001 NTU. Its accuracy is ±2% of reading, or ±0.015 NTU, whichever is greater, and ±5% of reading above 40 NTU. Its step response time is 90% in 120 sec. Like its raw-water counterpart, turbidity and temperature are pre-calibrated, and the sensor provides validation information. The drinking-water sensor has specific power requirements that are higher than other Datasticks: 24 VDC at 8 watts, with local display and flow through cell.

All Datastick sensor assemblies are CE-marked for emissions/immunity and safety, and are UL-rated Class I, Division 2. **Aquasensors LLC, www.aquasensors.com (262) 255-4459.**

VIDEOGRAPHIC RECORDERS BECOME TINY POWERHOUSES



This four-channel, 4-in., strip chart recorder replacement is called a videographic recorder, but it's much smaller and hardier than any other videographic recorder we've ever seen. ABB's SM500F is available in wall, pipe, or panel-mounting with the same NEMA 4X enclosure, and can be supplied with a "daylight bright," TFT color, LCD screen. The unit can be used to display trends like a strip chart recorder, or as a multiple-indicator display, or both, from up to eight calculated channels (four analog/digital inputs, math results, or Modbus TCP data). The unit has a large flash memory for recording, and automatically archives data onto commercially available SD flash memory cards. The unit can be supplied with up to three relay outputs, 32 process alarms, four real-time alarms, two custom linearizers, up to 16 software totalizers and 8 math functions, as well as Ethernet communications. Software provides a built-in audit trail for CFR21 Part11 and Sarbanes-Oxley compliance. **ABB; 215/674-6000; www.abb.com**

CORIOLIS MASS FLOWMETER GETS TOUGH

Enhancements to the Quantim Coriolis meter and controller include a Hastelloy C-22 sensing tube for use



in environments encountering corrosive fluids, and a high-pressure option that can be specified for use in applications with pipeline pressures up to 4,500 psi. More info at controlglobal.com. **Brooks Instruments; 215/362-3608; www.emersonprocess.com/brooks**

HEAVY-DUTY CONNECTORS

Mini-HMC heavy-duty connectors have a hard shell, but are 65% smaller than the company's previous switches, making them suitable for applications such as small robots, industrial controls, factory automation, and produc-



tion equipment. The connectors have a crescent-shaped metal hood with single-action lock and handle up to 7A. More info at controlglobal.com. **Molex; 630/527-4815; www.molex.com**

ETHERNET/IP SWITCHES

Industrial-grade EISXM, EICPM and EISBM EtherNet/IP managed switches have full-duplex on all ports, IGMP snooping, port mirroring, VLAN, auto-negotiation, manually configurable speed/duplex, wire-speed switching fabric, SNMP for switch management, and IEEE 802.1D Spanning Tree Protocol (STP). The switches meet all the requirements for Rockwell Automation's EtherNet/IP. More info at controlglobal.com. **Contemporary Controls; 630/963-7070, x102; www.ccontrols.com**

FLOW/LEVEL/TEMPERATURE SWITCH

FlexSwitch FLT Series flow and level switch provides monitoring, controlling and alarming on flow rates or levels of critical fluids such as water, air,



gases, coolants and lubricants used in metals and material processing and equipment. It's available in insertion and in-line styles for either pipe or tube installation, is suitable for use in ATEX locations, and is rated for NEMA Type 4X (IP66) environments. More info at controlglobal.com. **Fluid Components International; 800/854-1993; www.fluid-components.com**

256 CHANNELS, BUT NOTHING'S ON

7140-430 software radio transceiver provides 256 individually tunable receive channels in a full-function PMC/XMC transceiver module. This combination is suitable for developers of multi-channel communication systems, such as wireless base stations that require a high channel density within tight size, weight, cost and power constraints. More info at controlglobal.com. **Pentek; 201/818-5900, ext. 767; www.pentek.com**

ACTUATOR ACTUALLY MOVES AXIALLY

The slide unit remains stationary while the profile moves axially in the DGEA cantilever electric linear actuator. Suitable for space-contained multi-axis configurations, the motor and cables remain fixed, minimizing the moving mass to just the load and the aluminum profile. It comes in two mounting configurations: with an in-line coupling or with an integrated, right-angle gearbox. More info at controlglobal.com. **Festo; 800/99-FESTO; www.festo.com**

SAFETY LIMIT SWITCHES

SL Safety Limit Series interlock switch is available with a large selection of actuator configurations with either plastic or die-cast housings. A rotating



head provides four actuator mounting positions. A NEMA 4 enclosure allows the switch to withstand water washdown cleaning. Contact configurations available include two to four sets of contacts with snap action, break before make, and make before break operating versions. **Scientific Technologies; 800/479-3658; www.sti.com**

SECURE SERVERS SERVE OLD AND NEW

The Secure Device Server line can communicate with any new device based on embedded Windows operating systems, plus any device with an RS-232/422/485 serial port or Ethernet interface, including 10 to 20 year-old legacy serial devices such as meters, SCADA appliances, PLCs and sensors. The server offers secure local and remote VNC, remote desktop, HTTP and telnet access over the network or serial lines. More info at controlglobal.com. **Opengear; 801/282-1387; www.opengear.com**



REAL-TIME NETWORK

NDDS real-time publish-subscribe communication middleware can be used in low-cost, memory-constrained applications. It requires less than 100 kilobytes of memory, works with programmable logic controllers (PLCs) and smart sensors, and supports the



Object Management Group's (OMG) Data Distribution Service (DDS) standard. More info at controlglobal.com. **Real-Time Innovations; 408/200-4700; www.rti.com**

HYGIENIC CORIOLIS METER

H-Series Coriolis meter has a 15 Ra electro-polished surface finish and is drainable, making it suitable for hygienic, clean-in-place (CIP) applications. It is certified by the European Hygienic Equipment Design Group (EHEDG), 3-A authorized, and constructed according to ASME guidelines for bio-processing equipment. The meter is constructed from 100% 316L stainless-steel wetted parts for cleanability and chemical compatibility with process fluids and cleaning solutions. **MicroMotion; 303/527-5301; www.EmersonProcess.com/micromotion**

SLICK ENHANCMENTS

Enhancments to Allen-Bradley SLC 500-Series Controllers include improved messaging over Common Industrial Protocol (CIP) networks, and enhanced security capabilities in control system programming. Instruc-



tion updates allow generic CIP commands to be initiated in devices, such as drives, that communicate on EtherNet/IP, ControlNet and DeviceNet networks. Firmware includes a secured processor capability through the company's RSAssetSecurity. More info at controlglobal.com. **Rockwell Automation; 800/223-5354, x1891; www.rockwellautomation.com.**

HISTORIAN SUPPORTS SQL

IndustrialSQL Server historian supports Microsoft SQL Server 2005 data management and analysis software. The Common Language Routine (CLR) function offers a programming environment which enables customers to use C++, Visual Basic.NET and other programming languages and tools to create custom applications. More info at controlglobal.com. **Wonderware; 949/459-9696, x244; www.wonderware.com**

NO I/O IS AN ISLAND

Advantys OTB distributed input/output system includes three communication bases for various types of fieldbus, CANopen, Ethernet TCP/IP or Modbus RS 485 serial. Users can create I/O islands managed by a master controller via a fieldbus or communication network. Each communication base includes 20 I/O, totaling 12 to 24 VDC inputs, six relay outputs and two to 24 VDC solid-state outputs. Expansion and interface modules clip on to a 35-mm symmetrical rail and enable configurations of up to 132 digital I/O and up to 48 analog I/O channels, or a mixture of both. **Schneider Electric; 800/392-8781; www.us.schneider-electric.com**



CONDUIT ADAPTERS QUICKLY DISCONNECT

Quick-disconnect conduit adapters allow fast transitioning from conduit to 7/8-16UN minifast, M12 eurofast and M23 multifast connectors. The front of the adapter provides quick disconnect compatibility, while the back provides screw terminals for wiring inside the conduit. The conduit adapters are available with one or two connectors, and attach to standard conduit bodies. **Turck; 800/544-7769; www.turck.com**



LASER SENSOR SEES 10 METERS

World-Beam QS18LD laser-diffuse and QS18LLP laser-retroreflective sensors use visible Class 1 lasers to see targets up to 300 mm or 10 m, respectively. Six connector options are available: 2 m or 9 m integral cable; four-pin Euro-style or integral pigtail QD; and four-pin Pico-style or integral pigtail QD. More info at controlglobal.com. **Banner Engineering; 888/373-6767; www.bannerengineering.com**



FIELDBUS POWER SYSTEM

F800 series redundant, fieldbus power supply provides eight channels of power for fieldbus installations. A network monitoring option continuously checks the health of installed fieldbus segments by measuring segment voltages, signal levels of all devices and noise in three bands. Fieldbus-to-shield short circuits and improperly terminated wiring are indicated. More info at controlglobal.com. **MTL Instruments Group; 603/926-0090; www.mtl-fieldbus.com**

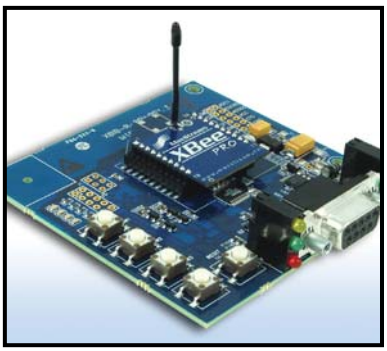
RADAR LEVEL GAUGE

Optiwave two-wire radar level gauge has a purge system that can be used in any application where gas, liquid, paste or solids can stick or cake, causing the antenna to be fouled with production adhesions or drying residue. The system cleans the entire antenna by purging the Teflon impedance transformer, the waveguide, and the antenna simultaneously. More info at controlglobal.com. **Krohne; 800/flowing; www.krohne.com**



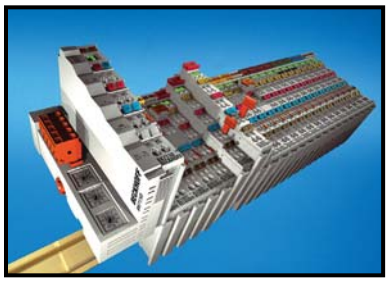
ZIGBEE DEVELOPMENT KITS

XBee professional developer's kit is suitable for developers working with mesh networking capabilities of the ZigBee standard. XBee-PRO starter kit and XBee starter kit make it possible to establish an RF link in minutes. All the necessary cables and testing components are included with each kit, and have options for loading 802.15.4, ZigBee or Freescale SMAC software. More info at controlglobal.com. **MaxStream; 866/765-9885; www.maxstream.net**



I/O BUS COUPLER

BK7150 bus couplers connect DeviceNet, Profibus, Ethernet TCP/IP, SERCOS, ControlNet, CANopen,



Lightbus, and CC-Link networks to the company's I/O. The coupler recognizes connected I/O terminals and automatically generates the configuration and affiliations of all inputs and outputs to the system's process image. More info at controlglobal.com. **Beckhoff Automation; 952/890-0000; www.beckhoffautomation.com**

CUSTOM CONTROL PANELS

Custom-engineered control panels can be integrated and manufactured



to match OEM or contractor project requirements. Panels come with human machine interfaces (HMI), appropriate displays, enclosures, mounting arms or devices, and all electrical and pneumatic fittings. More info at controlglobal.com. **Metfab Technologies; 401/732-6677; www.metfabtech.com**

CONTROL SNAPSHOT

The Loop Performance Snapshot in PlantTriage software includes graphical displays of critical loop-assessment information, including oscillation detection, comparison of key performance indicators (KPIs), linearity and valve hysteresis assessment, variability, time in normal, and dynamic process models. Red, yellow, and green color codes draw users' attention to problem areas. More info at controlglobal.com. **ExperTune; 262/369-7711; www.expertune.com**

PROGRAMMER/CONTROLLER UPGRADE

The latest version of IPC5000 universal controller/programmer includes two loops, accuracy of 0.10%, and storage for up to 32 programs with a total of 800 segments. It has two universal



analog inputs, two analog outputs, 12 digital inputs, and 12 digital outputs. Applications include furnace control, environmental chambers, laboratory ovens, sterilizers, chemical batching, dryers and dyeing machines. More info at controlglobal.com. **Honeywell; 215/641-3798; www.honeywell.com/ps**

FAST CPUS FOR PLCS

The FA-M3R family of Multi-Controller PLC systems have been upgraded with S-series CPUs that have a minimum instruction execution time of 0.07 microseconds. All automatically store program documentation, such as rung comments, I/O description and user tag names, locally in CPU protected memory. This allows for much



easier maintenance and program development. **Yokogawa; 800/258-2552; www.yokogawa.com**

REPEAT THAT, PLEASE

A broadcast repeater mode is available with XTend and XStream RF modems. Activating the repeater mode extends the range of the network by allowing



each node to receive and forward all broadcast transmissions, which allows nodes outside direct RF range to communicate. More info at controlglobal.com. **Maxstream; 866/765-9885; www.maxstream.net**

CONTROL SOFTWARE SEES NETWORKS

Enhancements to TwinCAT 2.10 software include functions for scanning



hardware connected to systems via Profibus, CANopen, DeviceNet, SERCOS, real-time Ethernet, Lightbus and EtherCAT. It can scan networks and find devices, compare networks to system configurations to identify what hardware is physically present, and synchronize multiple deterministic networks. More info at controlglobal.com. **Beckhoff Automation; 952/890-0000; www.beckhoffautomation.com**

All Wound Up



Greg McMillan and Stan Weiner, PE, bring their wits and more than 66 years of process control experience to bear on your questions, comments, and problems. Write to them at controltalk@putman.net.

Greg: We feel like we've neglected our motor enthusiasts, so this month we present an interesting case of "now you see it, now you don't" in a pump system submitted by George Bentinck, Chip Scavone, and Dave Schnelle. The system consists of pressure controllers at the inlet and outlet of parallel solution filters as shown in Figure 1. The inlet and outlet pressures are controlled by the manipulation of DC drives on individual stuffing gear pumps and metering gear pumps, respectively.

George: A notable pressure limit cycle (~35% of scale at about a 30-sec period) developed in the two pressures, causing periodic process shutdowns or poor downstream product quality. Attempts at tuning (on-line and via simulation) and minimizing loop dead time by speeding up control logic execution yielded little improvement. The waveform appearance is similar to that seen with a "sticky" control valve. However, this process doesn't use control valves; it uses DC variable-speed drives and speed-reducing gearboxes between the drive and gear pump. Drive tuning and performance were checked by a corporate expert, and found to be okay. Acoustical analysis of gearboxes didn't indicate defects. After months of living with this problem, a pump motor developed a bad winding, and was replaced. Pressure variability was greatly reduced to normal levels when the system was restarted with the new motor!

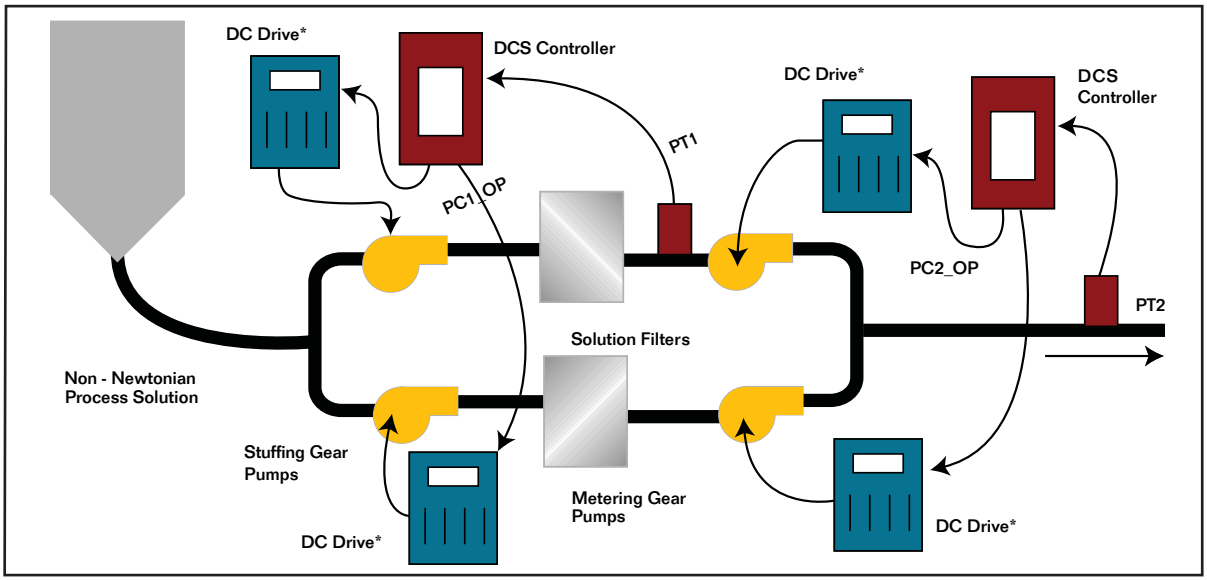
Greg: A positive-displacement pump is a constant flow device, which means the pump curve is steep and process gain is high (small variation in flow causes a large variation in pressure). There's a limit cycle with an amplitude that's proportional to the resolution limit of the drive (e.g. 0.2%) multiplied by the process gain. This, combined with the interaction between pumps in series and parallel, could lead to bigger oscillations. Beyond checking the A/D resolution of the input cards, I'm a bit out of my element here (final element humor), so I asked Bob Heider, who has much more experience in the ins and outs of motors, to take a look at the application.

Bob: I simulated the pump response for three different windings of DC mo-



FIGURE 1.

PUMP SYSTEM



This interacting pressure control system produces different pump responses.

tors to a step change in viscosity in a non-Newtonian fluid. For "separated excited" winding, the peak change in flow was only 0.1 gpm, versus 0.8 gpm for a "shunt" winding, and 1.8 gpm for a "series" winding.

Greg: Windings of the wrong type or in bad condition could increase the effect of a process disturbance or load change to the point where it triggers sustained oscillations from resolution limits and interactions. Before we get all wound up, let's break for a twist and a turn by Stan.

Stan: We've been criticized for unintentionally neglecting the mechanical aspects of control. This is probably because of our backgrounds and experience. However, we've been approached to consult with the "Work 4 Co," which is developing sensors and controllers to properly tighten tops or lids on bottles, jars and other containers.

The question/problem is: "How Tight Should Tops Be Tightened (HTSTBT)?"

If lids are too tight, senior citizens, the handicapped, people with arthritis, and others with weak fingers can find it extremely difficult or impossible to open some containers. Some people suggest not tightening tops so tightly, but manufacturers respond that not tightening them enough will cause leakage and spoilage, and allow more children to accidentally open bottles and jars".

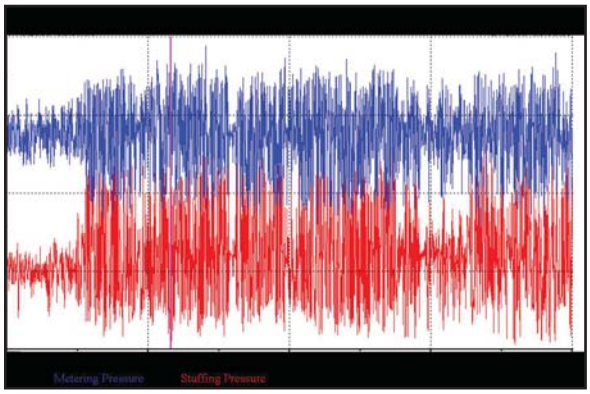
Consequently, we'd like to help Work 4 design and build a Variable-Tightness Top Machine (VTTM). This is our initial design concept: as customers enter a market, they go to a panel on the wall, enter their secret code, and turn a large dial that would look like a typical container top. This Variable Torque Dial (VTD) would measure their hand strength, and become the set point for the VTT Controller (VTTC). When their shopping is done, these customers would go to the checkout counter, and identify themselves by either entering their secret code or inserting a credit card. The cashier will scan the items for pricing as usual, and then the cashier, bagger, or customer will pass the jars and bottles through a Variable Torque Torguer (VTTR). This VTTR, in conjunction with the VTD and VTTC, will provide a VTT.

Another problem: after the VTT is set, the sack packer or bagger can't pick up the container by the top, which they have a tendency to do. Picking up the item by the top may negate the setting on the VTTR.

"Editor:" Since Greg has his time split playing consultant and going on vacation; we'll need some kind of control algorithm for the VTTC. Stan is occupied adjusting the pH and the free and combined chlorine concentration of his salt pool. So don't hes-

FIGURE 2.

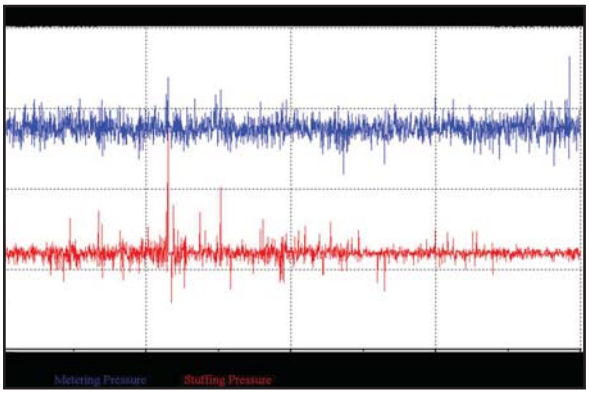
PRESSURES BEFORE MOTOR REPLACEMENT



Metering and stuffing pressures before pump motor replacement (Y-axis scale = 80% of Span)

FIGURE 3.

PRESSURES AFTER MOTOR REPLACEMENT



The same variables, on the same scale, after replacement of the metering pump motor.

itate. Send in your educated guess. The winner will be invited to attend the first market to successfully offer this boon to weaker hands: custom-torqued tops!

Greg: We top this column off with a Top 10 list from Mark Sowell and a puzzler from Hunter Vegas.

Top 10 People You'd Like to See in the Dunking Booth at Your Company Picnic

10. The IT security person responsible for your company's password policy.
9. The environmental representative on your project.
8. The accountant that won't let you use the current cost of natural gas to justify your project.
7. The department head that requires everyone to fill out a monthly timesheet.
6. The designer of your company's automated expense reporting system.
5. The process engineer that saved a buck by putting an actuator on a block valve.
4. The operator that calls you in to tune a loop whose output is saturated.
3. The project manager that notifies you that the configuration work you were asked about last year needs to be downloaded tomorrow.
2. The manufacturing engineer that blames the spill on your multivariable controller that wasn't in service because the operator had everything in manual.
1. The engineer rolling out the new corporate interlock guidelines.

Puzzler by Vegas

Hunter: In this puzzler, a plant installed large, 4-in. flange, integral, pad-type DP level transmitters on several tanks. Each DP meter used a 25-ft capillary and a 2-in. flange pressure pad on the low side of the meter to compensate for tank-vapor pressure changes. The plant had to constantly recalibrate the zero on the meters every time they emptied the tanks, and were considering installing a different manufacturer's transmitters to eliminate the problem. Will wholesale transmitter replacement solve their problem?

Next Month: Answers

In the July 2006 issue of *Control*, we'll compile and answer the last six months-worth of puzzlers, and maybe we'll all learn something new. And, as usual, we'll have another top 10 list. If you have any suggestions for top 10 lists, by all means let us know.

This Month's Puzzler: Shifting the Zero?

Will wholesale transmitter replacement solve the problem? Or is the plant stuck with constantly recalibrating the zero every time the tank empties? Do ***you*** know the answer?

Send an e-mail to controltalk@putman.net with your answer to the Puzzler, or send in your questions or comments.

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Connectivity That Counts

I think I was 36,000 feet over Greenland when it struck me. One of the last legitimate bastions of coveted disconnectedness from the rat race of electronic communication was under assault—and I found myself in full collusion.

You see, time was when a long flight, especially a transoceanic one, provided the perfect venue for Stephen Covey's crucial, but often-neglected, Quadrant-2 activities, which he famously defined as "not urgent but important." But now, seduced by Lufthansa's in-flight Wi-Fi connection on the way back home to Chicago from Frankfurt, only the relatively nominal $27 fee separated me from several bonus hours of "productive" connectedness. And I answered the siren's call—hook, line, and sinker.



Pretty soon the guy next to you will be Skyping away, and even your brief magazine read or nap will be shot.

I don't mean to imply that connectivity in itself is bad. Indeed, it's indispensable. It's just that lately we seem to have all too much of the wrong kind. Or, more precisely, we have the kind that doesn't go far enough. Connectivity is easy; it's meaningful communication that's difficult.

From an industry perspective, I think the process automation parallels are evident. Twenty years ago, systems were easily connected, but *meaningful* integration was hard. It's as if one system spoke French, the other English. So, while they could hear each other, they couldn't understand. Today, much fruitful activity in the process automation realm is on the meaningful integration of applications with each other, so that true knowledge and information—and not just more data—is the outcome.

Many technical professionals tend to focus on specifications and other "facts," but the more meaningful communication that comes with face-to-face interaction and a handshake is how trust is built and how truly significant things are accomplished. E-mail and the other electronic means of communication will never rival face-to-face events, where people exchange information and develop relationships that could never coalesce over the web.

Two quick examples: first, last week's travels took me to the beautiful Canadian Rockies (no in-transit Wi-Fi connection, this time!) and Matrikon's Valued Partner event in Banff, Alberta. Here, the majority of presentations to the group, from Matrikon's management and other key partner-companies, dwelt not on technology as primary challenge, but on issues of people and company culture.

Meanwhile, the latest *AutomationXchange*, in Carefree, Ariz., (disclosure: hosted by *Control*'s sister publication *Control Design*) brought together OEM engineering teams with suppliers of industrial automation technology in an unprecedented, one-on-one exchange of ideas and solutions.

The key to *AutomationXchange*'s success? An investment by both parties before the event to understand the potential fit for a business relationship, which is then explored and developed in person at *AutomationXchange*.

For me, the qualitative difference between electronic connectivity and personal communication is driven home by a story once told to me by a good friend and church pastor, Tom Hickey. He tells the tale of a young child, who, having been told by her parents how Jesus and his angels will always be with her to protect her, nevertheless makes repeated appearances in her parents' bedroom whenever a thunderstorm strikes. "Don't you know God will protect you?" asks her mother. To which the child replies, "I know that in my head and in my heart, but sometimes I just need some skin."

And so it is with us. When it comes to solving complex engineering and manufacturing problems–e-mails, web searches, and solo intellectual firepower often will only get you so far. Take the time to forge meaningful relationships with colleagues, with suppliers, and with your customers–be they inside or external to your own organization.

Then, the next time you're on a long flight, and are tempted to see what's new at the office, resist the temptation, I urge you! Spend some guilt-free time in Quadrant 2, preparing yourself to more effectively communicate when you're next face-to-face with your business associates.

Because, if we're not careful, pretty soon the guy next to you will be Skyping away about delivery times and the office holiday party, and even your brief magazine read or nap will be shot. Next up, of course, will be a planeload of simultaneous, mobile-phone connections. Me? I'll take the crying baby in seat 12-D.

Keith Larson,
VP content, Putman Media

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Wonderware Touch Panel Computer (on left).

Wonderware Industrial Tablet (docking unit not shown).

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